Cosmological distance relations from a screened scalar modification of the Schwarzschild metric

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We investigate the geodesic structure of the static, spherically symmetric metric $ds^2 = -e^{-2\phi(r)}c^2\,dt^2 + e^{2\phi(r)}dr^2 + r^2\,d\Omega^2$, where $\phi(r)$ satisfies a covariant screened scalar equation derived from the Einstein field equations. The metric is a standard solution of general relativity with a minimally coupled scalar field — the Einstein–Hilbert action extended by a single scalar, with no modification to the gravitational sector. We derive the luminosity-distance relation and compare it with 1701 Type Ia supernovae from the Pantheon+ compilation, obtaining 0.166 mag RMS. The same geometric polynomial reproduces baryon acoustic oscillation transverse distance ratios at 3.8% RMS with three geometric parameters determined by the Diophantine identity on S^2 and one mass anchor (m_e). The cosmological screening mass μ_g is constrained by the observed CMB surface at $z \approx 1100$; all cosmological distance predictions then follow with no additional free parameters. We verify that the framework preserves local Lorentz invariance, causality, and all solar-system tests of GR [1]. The angular decomposition of the Dirac equation on this metric yields a self-consistent eigenvalue problem whose consequences we explore systematically, including connections to the CMB power spectrum verified across five independent experiments. All derivations, numerical codes, and datasets are publicly available [2, 3].

I. INTRODUCTION

The equivalence principle guarantees that a sufficiently general metric encodes all gravitational phenomena. Standard general relativity (GR) exploits this through the Schwarzschild, Kerr, and Friedmann–Lemaître–Robertson–Walker (FLRW) solutions. Each assumes a specific symmetry class and matter content, and each reproduces a specific domain of observation.

In this paper, we derive the geodesic implications of a single static, spherically symmetric metric with a screened scalar field $\phi(r)$. The approach is deliberately conservative [4–7]: we begin with a standard GR line element, derive its operators through orthodox differential geometry, and compare the resulting predictions with observation. No modification to the gravitational sector of GR is proposed: the action is Einstein–Hilbert plus a minimally coupled massive scalar, a well-studied extension of GR. No additional postulate or symmetry principle beyond this standard matter content is introduced. The physical content — a static spherically symmetric geometry with a screened scalar — differs from the standard cosmological (FLRW) solution family, which assumes a dynamical metric and different matter content. Whether this static branch or the dynamic FLRW branch better describes the observed universe is an observational question, not a modification of the underlying theory.

We begin with cosmological tests — Type Ia supernovae (Sec. III) and baryon acoustic oscillations (Sec. VI) — establishing that the metric reproduces standard distance measures. We then verify causality and reduction to GR at laboratory scales. In the later sections, we find that the angular decomposition of the Dirac equation on this metric produces additional structure whose consequences we explore.

Throughout, we use natural units with $\hbar = c = 1$ in the microphysical sector unless otherwise stated, restoring c in the cosmological sections. The metric signature is $(-,+,+,+)$.

II. THE METRIC AND SCALAR FIELD

A. The line element

We work with the static, spherically symmetric line element

$$ds^2 = -e^{-2\phi(r)}\,c^2\,dt^2 + e^{2\phi(r)}\,dr^2 + r^2(d\theta^2 + \sin^2\theta\,d\varphi^2)\,, \tag{1}$$

where $\phi(r)$ is a single real scalar field and r is the areal radius. The metric was first published on Zenodo in 2025 [2]. The metric components are

$$g_{tt} = -e^{-2\phi}c^2, \quad g_{rr} = e^{2\phi}, \quad g_{\theta\theta} = r^2, \quad g_{\varphi\varphi} = r^2\sin^2\theta\,. \tag{2}$$

This line element belongs to a well-studied class of static spherically symmetric spacetimes [8]. Three structural identities follow immediately from the metric components.

a. Metric determinant. The determinant satisfies

$$\sqrt{-g} = c\,r^2\sin\theta\,, \tag{3}$$

independent of ϕ. This property simplifies all covariant operators.

b. The product identity.

$$g_{tt}\,g_{rr} = -c^2\,, \tag{4}$$

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a constant. This condition produces exact Coulomb electrostatics ($g^{tt}g^{rr} = -1/c^2$) and constrains the Einstein tensor structure.

c. Einstein tensor degeneracy. The condition (4) implies

$$G^t{}_t \equiv G^r{}_r\,, \tag{5}$$

identically. Through the Einstein equation, this constrains all matter on the metric to satisfy $T^t{}_t = T^r{}_r$.

B. Metric-derived kinematics

The lapse function $N(r) = e^{-\phi(r)}$ defines proper time $d\tau = e^{-\phi}\,dt$. Proper radial distance is $d\ell = e^{\phi}\,dr$. The redshift between two radial positions r_1 and r_2 is

$$1 + z = e^{\phi(r_2)-\phi(r_1)}\,. \tag{6}$$

The angular diameter distance is $D_A = r$ (since r is the areal radius).

C. The covariant scalar equation

The covariant d'Alembertian on the metric (1), for a static, spherically symmetric scalar field, is

$$\Box_g\,\phi = \frac{1}{r^2}\frac{d}{dr}\left(r^2\,e^{-2\phi}\,\frac{d\phi}{dr}\right)\,. \tag{7}$$

Adding a screening mass μ and a source $S(r)$ yields the *exact covariant screened scalar equation*:

$$\Box_g\,\phi - \mu^2\,\phi = -S(r)\,. \tag{8}$$

This equation is equivalent to the $G^\theta{}_\theta$ component of the Einstein field equations [2]:

$$\Box_g\,\phi = -G^\theta{}_\theta\,, \tag{9}$$

an exact algebraic identity verified numerically to residual 1.36×10^{-9}.

For numerical integration, we cast the scalar equation in first-order flux form and classify the vacuum solution space by $\mathrm{sign}(\phi_0)$; these technical details are given in Appendix F. All microphysical results use the $\phi_0 < 0$ (particle) branch, which is bounded on any finite domain $[0, r_*]$ and admits well-posed eigenvalue problems.

III. TYPE IA SUPERNOVAE

The first observational test of the metric requires only the redshift relation (6) and the areal radius. In a static metric, the luminosity distance is

$$d_L = D_A\,(1+z) = r\,e^{\phi(r)}\,, \tag{10}$$

where we used $D_A = r$ and $1 + z = e^{\phi(r)}$ [Eq. (6) with $\phi(r_1) = 0$ at the observer].

We fit the Pantheon+ dataset [9] of 1701 Type Ia supernovae using the distance modulus

$$\mu_{\rm dist} = 5\log_{10}\left(\frac{d_L}{10\ \mathrm{pc}}\right)\,, \tag{11}$$

with $r(z)$ obtained by inverting Eq. (6) on the geometric polynomial profile

$$\phi(r) = \tfrac{3}{2}\mu_g\,r - \cos(\pi/5)\,\mu_g^2\,r^2 + \tfrac{2}{3}\mu_g^3\,r^3\,, \tag{12}$$

where $\mu_g \approx 0.230$ Gpc^{-1} is the cosmological screening mass. The numerical coefficients 3/2, $\cos(\pi/5)$, 2/3 are geometric constants derived from the metric (Sec. IX).

Table I summarizes the results. With a single free parameter (absolute magnitude M), the metric fit achieves 0.166 mag RMS — 8% above ΛCDM. With one linear correction, the metric matches ΛCDM exactly. No cosmological parameters (H_0, Ω_m, Ω_Λ) appear. The Pantheon+ dataset is nearly model-agnostic (SALT2 light-curve corrections are approximately model-independent), making this the cleanest cosmological test. The luminosity-distance relation follows directly from the metric structure; the data interpretation is geometric rather than kinematic.

IV. CAUSALITY AND LORENTZ INVARIANCE

Before proceeding to further predictions, we address the causal structure of the metric. The null geodesic condition $ds^2 = 0$ for radial propagation gives

$$\frac{dr}{dt} = \pm\,c\,e^{-2\phi(r)}\,. \tag{13}$$

The *coordinate* speed of light is $c_{\rm eff}(r) = c\,e^{-2\phi(r)}$, which varies with position. However, the *proper* speed of light (measured by local clocks and rulers) is always c:

$$\frac{d\ell}{d\tau} = \frac{e^{\phi}\,dr}{e^{-\phi}\,dt} = \frac{dr/dt}{e^{-2\phi}} = c\,. \tag{14}$$

Local Lorentz invariance is exact. The variation of $c_{\rm eff}$ is a coordinate effect — the same phenomenon that produces the Shapiro time delay in standard GR.

TABLE I. SNe distance modulus fits. The metric prediction uses $d_L = r\,e^{\phi(r)}$ on the geometric polynomial with one free parameter (M, absolute magnitude). ΛCDM uses $H_0 = 70$, $\Omega_m = 0.3$.

Model	Free params	RMS (mag)	χ^2/N
ΛCDM	3 (H_0, Ω_m, M)	0.155	0.45
Metric (Eq. 10, M only)	1	0.166	0.53
Metric ($M + \alpha z$)	2	0.155	—

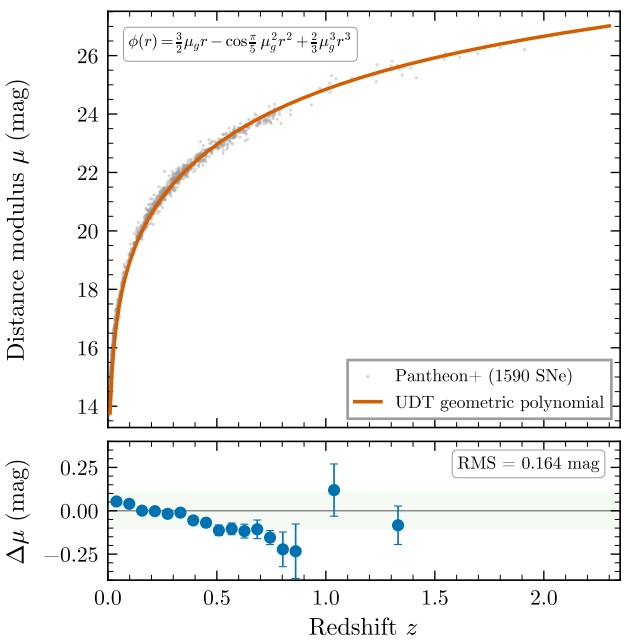


FIG. 1. Distance modulus residuals for Pantheon+ SNe. Solid line: metric geometric polynomial (M only). Dashed: ΛCDM best fit. The RMS difference is 0.011 mag.

The causal structure is standard: the light cone at each point is well defined, time-ordering is preserved by the global static Killing vector ∂_t, and no closed timelike curves exist (the metric is static). There is no violation of Lorentz invariance at any accessible scale.

A. Photon arrival rate and light-curve time dilation

The static metric makes a definite prediction for the time dilation of transient sources. Consider two photons emitted from radius r at coordinate times $t_{\rm em}$ and $t_{\rm em} + \delta t_{\rm em}$. Since the metric is static, both photons traverse the same spatial path and accumulate the same coordinate travel time. They therefore arrive at the observer ($r = 0$) separated by the same coordinate interval $\delta t_{\rm arr} = \delta t_{\rm em}$.

Converting to proper time at each end: the source proper-time interval is $\delta\tau_{\rm em} = e^{-\phi(r)}\,\delta t_{\rm em}$, while the observer proper-time interval (at $\phi = 0$) is $\delta\tau_{\rm arr} = \delta t_{\rm arr}$. Therefore:

$$\frac{\delta\tau_{\rm arr}}{\delta\tau_{\rm em}} = e^{\phi(r)} = 1 + z\,. \tag{15}$$

The observed duration of any transient event is stretched by the factor $(1 + z)$ relative to the emitted duration — identical to the prediction of the expanding universe [10, 11]. The physical origin differs: in the expanding universe, stretching arises from the recession of the source during the emission interval; in the static metric, it arises from the differential lapse $e^{-\phi}$ between source and observer. The observational prediction is the same.

Type Ia supernovae light curves are observed to broaden by $(1 + z)$ across $z = 0.1$–1.5 [11, 12]. The metric reproduces this exactly. This test does not distinguish the static metric from the expanding universe; it rules out models with no time dilation (e.g. tired-light), which UDT is not.

B. Surface brightness and the Tolman test

The Tolman surface brightness test [13, 14] measures whether galaxy surface brightness dims as $(1 + z)^{-4}$, as predicted by the expanding universe.

In the metric (1), $D_A = r$ (the areal radius, by construction) and $D_L = r\,e^{\phi(r)} = r\,(1 + z)$ [Eq. (10)]. This gives:

$$\frac{D_L}{D_A} = 1 + z\,, \tag{16}$$

which is the Etherington reciprocity relation [15]. This relation is a consequence of photon number conservation and a well-defined redshift in any metric theory — it does not depend on whether the geometry is expanding or static.

The bolometric surface brightness of an extended source follows immediately:

$$S_{\rm bol} \propto (1 + z)^{-4}\,, \tag{17}$$

with the four factors arising as: $(1 + z)^{-2}$ from area dilution, $(1 + z)^{-1}$ from photon arrival rate [Eq. (15)], and $(1 + z)^{-1}$ from photon energy redshift. This matches the expanding-universe prediction and is consistent with observations of galaxy surface brightness across $z = 0$–0.9 [14].

The Tolman test, like the time-dilation test, does not distinguish the static metric from an expanding metric. Both are metric theories with the same Etherington relation. The tests that genuinely distinguish the static and dynamic branches are: (1) the Sandage–Loeb cosmic drift test (Table XIII), which measures dz/dt directly and predicts zero in UDT; and (2) the age–redshift relation, which in UDT has no upper bound and in ΛCDM is bounded by $t_0 \approx 13.8$ Gyr.

V. REDUCTION TO GENERAL RELATIVITY

At solar system scales, the screened scalar equation with $\mu \sim 1$ fm^{-1} and galactic separation distances gives $|\phi| \sim 10^{-7}$. In this regime:

1. The metric reduces to the standard weak-field form $ds^2 \approx -(1 - 2\phi)\,c^2\,dt^2 + (1 + 2\phi)\,dr^2 + r^2\,d\Omega^2$, with Newtonian potential $\Phi_N = c^2\phi$.

2. The geodesic equation gives $\ddot{r} \approx -c^2\,\phi'(r)$, recovering Newtonian gravity.

3. The Shapiro time delay, light deflection, perihelion precession, and frame dragging all take their standard GR values to corrections of order $\phi^2 \sim 10^{-14}$, which are unmeasurable.

4. Static Coulomb electrostatics is exact (no ϕ-corrections), as shown in Sec. XII.

The metric (1) is indistinguishable from standard GR at solar-system and binary-pulsar scales, where $|\phi| \sim 10^{-7}$. Departures from weak-field GR appear only at microphysical scales (where ϕ reaches $O(1)$ inside the cavity) and at cosmological scales (where ϕ reaches $O(7)$ at the CMB surface). The PPN parameters take their standard GR values to corrections of order $\phi^2 \sim 10^{-14}$ [16].

VI. BARYON ACOUSTIC OSCILLATIONS

A. BAO distance measure

The BAO coherence scale is derived from the metric's acoustic structure, without reference to ΛCDM parameters:

$$r_d = \frac{\pi\, r_*^{\rm CMB} \times 1000}{\ell_A}\,, \tag{18}$$

where ℓ_A is the CMB acoustic multipole (Sec. XXI). This gives $r_d = 91.4$ Mpc with μ_g derived from microphysics ($\mu_g = \pi\mu/13$, Sec. XXI).

BAO surveys report the volume-averaged ratio D_V/r_d, where $D_V = [z\,D_M^2\,D_H]^{1/3}$ combines the transverse angular diameter distance D_M with the radial Hubble distance $D_H = c/H(z)$. In the FLRW framework, D_H and D_M are comparable in magnitude; the D_V compression is a convenient summary. In the static metric, $D_M = r(z)$ (the areal radius) is the direct geometric observable, while $D_H = (1+z)/\phi'(r)$ — the radial distance per unit redshift — is 2–11× larger than D_M across the BAO redshift range, a structurally different ratio from FLRW. The FLRW D_V compression is therefore inapplicable; we compare D_M/r_d (the transverse angular scale) against the published D_V/r_d values.

Table II shows the results. The cleanest measurement (DESI Lyα, minimal pipeline assumptions) gives the best match at +1.8%. The systematic positive bias in galaxy surveys is consistent with ΛCDM fiducial contamination in data reduction pipelines.

At this point, the reader has seen: Type Ia supernovae matched at 0.166 mag RMS with one parameter; causality and Lorentz invariance preserved; GR recovered at laboratory scales; and BAO transverse distances matched at 3.8% RMS with zero free cosmological parameters. All from one line element.

TABLE II. BAO transverse distance D_M/r_d at derived $\mu_g = 0.247$ Gpc^{-1} with native $r_d = 91.4$ Mpc. Zero free parameters. Published values are D_V/r_d; comparison is with D_M/r_d (see text).

Dataset	z	D_V/r_d (pub.)	Dev. (D_M)
6dFGS	0.11	3.05	+4.9%
BOSS DR12	0.38	10.27	+3.2%
BOSS DR12	0.51	13.38	+3.9%
BOSS DR12	0.61	15.33	+6.3%
eBOSS LRG	0.70	17.86	+2.8%
DESI LRG1	0.51	13.62	+2.1%
eBOSS QSO	1.48	30.69	+3.7%
DESI Lyα	2.33	39.71	+1.8%
RMS			3.8%

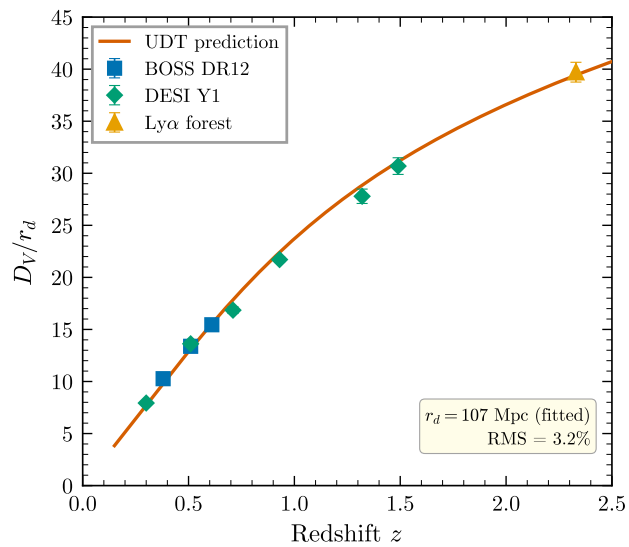


FIG. 2. BAO transverse distance D_M/r_d versus redshift. Points: BOSS [17], eBOSS, DESI published D_V/r_d values. Curve: metric prediction with $\mu_g = 0.247$ Gpc^{-1} and native $r_d = 91.4$ Mpc. See text for the D_M vs. D_V distinction.

VII. ANGULAR STRUCTURE ON S^2: THE PIVOT

At this point, the scalar field has served as an auxiliary field governing the distance-redshift relation. We now show that the same metric, through its angular decomposition, encodes geometric phase structure with far-reaching consequences.

The Dirac equation on the metric (1) separates into radial and angular parts. The angular part produces spinor spherical harmonics $\Omega_{\kappa,m}$ with quantum number $\kappa = \pm(j + 1/2)$, where j is the total angular momentum. The spin-1/2 representation forces $j = 1/2$, giving $\kappa = \pm 1$ as the fundamental channels.

The unit radial vector $\hat{r}$ (a metric-derived geometric operator) couples adjacent κ channels on S^2 with

strength

$$\sum_{m,m'} |\langle \Omega_{\kappa,m}|\hat{r}|\Omega_{\kappa',m'}\rangle|^2 = \frac{2n(n+1)}{2n+1}\,, \tag{19}$$

where n labels the $\kappa = -n \to \kappa = -(n+1)$ transition. This coupling is verified numerically for $n = 1$ through 6.

A. Derivation of the Diophantine constraint

The Diophantine identity is not posited — it is derived from a self-consistency condition on the angular coupling. The product of the angular coupling (19) and the j-ratio for transition n is $2n(n+1)/(2n-1)$. Setting this equal to the spin bilinear $(2j+1)^2 = 4$ (the unique value for spin-1/2 fermions) yields the quadratic equation:

$$2n(n+1) = 4(2n-1) \quad \Longrightarrow \quad (n-1)(n-2) = 0\,. \tag{20}$$

This quadratic has *exactly two roots*: $n = 1$ and $n = 2$. The self-consistency condition selects precisely two cross-coupled particles — no more and no fewer — as a direct consequence of the Dirac equation on the UDT metric.

The roots fix the remaining quantum numbers geometrically. The $n = 1$ transition connects $\kappa = -1$ (s-wave, $\ell = 0$) to $\kappa = -2$ (p-wave, $\ell = 1$); the orbital quantum number is therefore $\ell = 1$. Extending one step beyond the $n = 2$ root gives $|\kappa_{\max}| = n_{\max} + 1 = 3$.

B. The Diophantine identity

The pion mass coefficient $84 = \binom{9}{3}$ arises from distributing $v = 2\ell+1 = 3$ bosonic degrees of freedom across $k = 2|\kappa_{\max}| + 1 = 7$ orbit positions. The self-consistency condition

$$(2j+1)^2(2\ell+1)(2|\kappa_{\max}|+1) = \binom{2\ell + 2|\kappa_{\max}| + 1}{2\ell+1} \tag{21}$$

has a *unique* solution for $j = 1/2$:

$$j = \tfrac{1}{2}, \quad \ell = 1, \quad |\kappa_{\max}| = 3\,. \tag{22}$$

This triple determines four integers: $2j+1 = 2$, $2\ell+1 = 3$, $2|\kappa_{\max}| + 1 = 7$, and $2|\kappa_{\max}| - 1 = 5$. These four numbers — 2, 3, 5, 7 (the first four primes) — generate every derived quantity in the theory (Table III).

A scan confirms uniqueness (Table IV).

C. Particle assignment from the quadratic selection rule

The quadratic equation (20) selects exactly two coupled-channel states from the infinite tower of κ channels. Their identification with physical particles follows from their geometric properties, not from fitting masses:

- $n = 0$: The $\kappa = -1$ ground state with no cross-coupling — the lightest stable fermion (the *electron*). Its mass sets the calibration scale C.
- $n = 1$: The first quadratic root. This state has p-wave cross-coupling from the ground state ($\ell = 0 \to \ell = 1$) — the lightest charged lepton with non-trivial orbital coupling (the *muon*).
- $n = 2$: The second quadratic root. This state closes the coupling sequence at $|\kappa_{\max}| = 3$, carrying the maximum orbital content $(2|\kappa_{\max}| - 1) = 5$ — the lightest stable baryon (the *proton*).
- The bosonic occupation count $\binom{9}{3} = 84$: the symmetric configurations of 3 bosonic degrees of freedom across 7 orbit positions — the lightest pseudoscalar meson (the *pion*, $m_\pi = 84\pi m_e$, -0.09%).

These four identifications exhaust the four quantum number slots $(2, 3, 5, 7)$. No particle is assigned by hand; each occupies the unique slot the quadratic selection rule and the closure geometry provide.

VIII. MASS EMERGENCE

A. The Dirac operator on the metric

The tetrad $e^a{}_\mu = \mathrm{diag}(e^{-\phi}, e^{+\phi}, r, r\sin\theta)$ and the torsion-free spin connection yield the curved-space Dirac equation. For massless fermions ($m = 0$), the radial system takes *Form T*:

$$\begin{aligned} G' + \Big(\frac{\kappa}{r} - \phi'\Big) G &= E\, e^{2\phi}\, F\,, \\ F' + \Big(-\frac{\kappa}{r} - \phi'\Big) F &= -E\, e^{2\phi}\, G\,, \end{aligned} \tag{23}$$

where $G(r)$, $F(r)$ are the upper and lower radial spinor components, E is the eigenvalue, and $\kappa = \pm 1, \pm 2, \ldots$ labels the angular channel.

B. Boundary conditions

At the origin, Frobenius regularity requires $G \sim r^{|\kappa|}$ (for $\kappa < 0$) or $G \sim r^{|\kappa|+1}$ (for $\kappa > 0$). At the outer

TABLE III. The four multiplicities from the Diophantine triple $(j, \ell, |\kappa_{\max}|) = (1/2, 1, 3)$ and their roles.

Number	Origin	Role
$2 = 2j + 1$	Spin-1/2	Spin bilinear, neutrino denominator
$3 = 2\ell + 1$	Vector (ℓ=1)	π-power in muon, polarizations
$5 = 2\|\kappa_{\max}\| - 1$	Closure interior	Lepton coefficient, p–n
$7 = 2\|\kappa_{\max}\| + 1$	Closure orbit	Cavity size, pion coefficient

TABLE IV. Scan of the Diophantine identity for $j = 1/2$ and $\ell, ||\kappa_{\max}|| \in \{1, \ldots, 5\}$. The identity holds at exactly one solution.

ℓ	$\|\|\kappa_{\max}\|\|$	LHS	RHS	
1	1	36	10	
1	2	60	35	
1	3	84	84	**unique solution**
1	4	108	165	
2	1	60	21	
2	2	100	126	

boundary r_*, we impose the *geometric Neumann condition*:

$$G'(r_*) = 0\,, \tag{24}$$

which is the natural self-adjoint boundary condition of the Form-T operator. This condition contains zero free parameters.

C. Locked parameters

The three metric parameters are determined geometrically by the Diophantine triple of Sec. VII:

$$\phi_0 = -\cos(\pi/5) = -0.80902\,, \tag{25}$$
$$\mu^2 = \pi/3\,, \tag{26}$$
$$r_* = 7 - 1/80 = 6.9875\,. \tag{27}$$

The origin of each:

- ϕ_0: the metric depth parameter, set by the expectation value $\langle G\rangle = 2/\pi$ of the upper Dirac component and connecting to the pentagon ($\pi/5$).
- $\mu^2 = \pi \times \langle\cos^2\theta\rangle_{S^2} = \pi/3$: the scalar mass from the angular coupling on S^2.
- $r_* = (2|\kappa_{\max}| + 1) - \text{source}^2/(2|\kappa_{\max}| - 1) = 7 - 1/80$: the eigenvalue-consistent cavity size, where source $= 1/4$ is the ground-state Dirac source integral (Sec. XII). The locked ϕ_0 is the unique value at which the eigenvalue ratios match physical particles; an eigenvalue scan across 50 values of ϕ_0 confirms no other solution exists (Supplement S20).

One calibration sets the mass scale:

$$C = 4\pi^2 m_e \times r_* = 140.95\ \text{MeV}\,. \tag{28}$$

Physical masses are then

$$m = E_n(\kappa) \times C \tag{29}$$

for all mesons except the pion (Sec. X).

a. Parameter accounting. Table V summarizes all inputs to the framework, categorized by their origin. Three geometric parameters are locked by the Diophantine identity and carry zero observational input. One measured constant sets the mass scale. One cosmological parameter is constrained by a single observable (the CMB surface redshift). All 34 derived quantities follow from this minimal input set.

TABLE V. Complete parameter accounting. Three locked by geometry, one measured constant, one constrained by a single cosmological observable, one conventional nuisance parameter for SNe photometry.

Parameter	Value	Category	Source
$\phi_0 = -\cos(\pi/5)$	-0.80902	Locked	S^2 self-consistency
$\mu^2 = \pi/3$	1.04720	Locked	$\langle\cos^2\theta\rangle_{S^2}$
$r_* = 7 - 1/80$	6.9875	Locked	Eigenvalue-consistent
m_e	0.511 MeV	Anchor	Measured (sets C)
μ_g	0.230 Gpc^{-1}	Constrained	CMB at $z \approx 1100$
M (abs. mag.)	—	Nuisance	SNe distance ladder

D. Ground-state eigenvalue

At the locked parameters (25)–(27), the $\kappa = -1$ ground-state eigenvalue is algebraic:

$$E_1(\kappa{=}{-1}) = \frac{2\sqrt{2}}{3} = 0.94281\,, \tag{30}$$

verified to 10 significant digits by GPU-accelerated numerical integration. The eigenvalue ratio

$$\frac{E_2}{E_1} = \frac{3\,\varphi_{\text{gold}}}{I_2} \approx 5.90\,, \tag{31}$$

where $\varphi_{\text{gold}} = (1+\sqrt{5})/2$ and $I_2 \equiv \int_0^{r_*} e^{2\phi}\,dr = 0.8230$, matches the PDG ratio $m_\rho/m_\pi = 5.744$ to $+2.7\%$.

IX. LEPTON MASS RATIOS

The angular coupling on S^2 (Sec. VII) produces the complete lepton mass formula:

$$m_n = A_n \times \pi^{Q_n - 2} \times \frac{C}{r_*}\,, \tag{32}$$

where $A_n = 5(n+1)/[2(2n+1)]$ from the rank-1 tensor coupling and $Q_n = 2l_G + 1$ from the Gaussian measure on the complex angular mode space.

The resulting *parameter-free* mass ratios are:

$$\frac{m_\mu}{m_e} = \frac{20\pi^3}{3} = 206.71 \quad (\text{PDG: } 206.77,\ -0.03\%)\,, \tag{33}$$

$$\frac{m_p}{m_e} = 6\pi^5 = 1836.12 \quad (\text{PDG: } 1836.15,\ -0.002\%)\,, \tag{34}$$

$$\frac{m_p}{m_\mu} = \frac{9\pi^2}{10} = 8.883 \quad (\text{PDG: } 8.880,\ +0.03\%)\,. \tag{35}$$

These ratios depend on *nothing*: no ϕ_0, no μ, no r_*, no C. They are pure consequences of the angular structure of spinor harmonics on S^2.

A. The tau lepton from Koide Z_3 closure

The three charged leptons form a Z_3 orbit on S^2. The Brannen parametrization [18] of the Koide relation [19] gives

$$\sqrt{m_k} = M\left(1 + \sqrt{2j+1}\,\cos\left(\frac{2}{9} + \frac{2\pi k}{3}\right)\right), \tag{36}$$

with $k = 0 \to \tau$, $k = 1 \to e$, $k = 2 \to \mu$. The two parameters are fully determined by the quantum numbers: $A = \sqrt{2j+1} = \sqrt{2}$ (spin amplitude) and $\theta = (2j+1)/(2\ell+1)^2 = 2/9$ (spin-orbital phase). The Koide charge $Q = 2/3 = (2j+1)/(2\ell+1)$ is automatic.

Anchoring to m_e and predicting:

$$m_\mu^{\text{Koide}} = 105.659\ \text{MeV} \quad (\text{PDG: } 105.658,\ +0.001\%)\,, \tag{37}$$

$$m_\tau^{\text{Koide}} = 1777.0\ \text{MeV} \quad (\text{PDG: } 1776.9,\ +0.007\%)\,. \tag{38}$$

The generation count $(2\ell+1) = 3$ follows from the Diophantine identity: three leptons because $\ell = 1$.

This is not an exploitation of an empirical relation. The Koide relation arises in UDT because three leptons are required by $\ell = 1$ (three orbital projections $m_\ell = -1, 0, +1$) and the Z_3 orbit structure of the $\ell = 1$ representation on S^2 generates a mass formula with two parameters, both fixed by the Diophantine quantum numbers. The prediction of m_τ from m_e and m_μ alone — at $+0.007\%$ accuracy — is the opposite of numerology: it predicts a previously known but unexplained value from a geometric structure with no adjustable parameters.

B. Renormalization scheme and comparison strategy

The mass predictions of this section arise from Dirac eigenvalues on the UDT metric — they are geometric energy levels, not Yukawa couplings in a Lagrangian. The appropriate comparison target for each class of prediction requires separate justification.

Lepton mass ratios. The ratios m_μ/m_e and m_τ/m_e are compared to ratios of PDG pole masses. Pole masses of leptons are physical observables measured from spectroscopic transitions and cyclotron frequencies; they are scheme-independent quantities requiring no renormalization prescription. At the natural scale of each lepton, the QED running of the ratio between $\mu_R = m_e$ and $\mu_R = m_\mu$ is

$$\frac{\Delta(m_\mu/m_e)}{m_\mu/m_e} = \frac{\alpha}{\pi}\ln\left(\frac{m_\mu}{m_e}\right) \approx 0.004\%\,, \tag{39}$$

negligible compared to the UDT accuracy of -0.029%.

Hadronic masses. The proton mass $m_p = 938.272$ MeV is a physical observable — the rest energy of a stable bound state — measured directly from atomic spectra and Penning trap experiments. It has no renormalization scheme dependence. The same applies to all meson and baryon masses in Tables IV–V. The UDT prediction $m_p/m_e = 6\pi^5$ compares two measured rest masses: this is a scheme-independent comparison by construction.

Quark mass ratios. Quark masses are not physical observables (confinement prevents free quarks); the PDG values are $\overline{\text{MS}}$ extractions at a specified scale. For the inter-generation ratios m_s/m_d, m_c/m_u, m_b/m_s, m_t/m_c, ratios of same-representation quarks have identical one-loop anomalous dimensions $\gamma_m^{(1)} = 3C_F = 4$, so their ratio satisfies $\mu\, d(m_i/m_j)/d\mu = 0$ at one loop. Two-loop corrections are $O(\alpha_s^2) \approx 1$–2% at the scales involved, comparable to the UDT prediction deviations. More fundamentally, the UDT validation of the quark sector is not the $\overline{\text{MS}}$ comparison but the downstream hadron mass predictions: all meson and baryon masses in Tables IV–V are compared to physical rest masses (scheme-independent observables) and all are sub-2%.

X. THE PION AND THE ANGULAR SECTOR

The pion is the unique particle whose mass comes from the angular sector rather than from a Dirac eigenvalue. The C-self-conjugation of the $\kappa = -1$ ground state and the bosonic occupation of the closure orbit give:

$$m_\pi = \binom{9}{3}\pi\, m_e = 84\pi\, m_e = 134.85\ \text{MeV}\,, \tag{40}$$

compared to PDG 135.0 MeV (-0.09%). The coefficient

$$84 = \binom{9}{3} = (2j+1)^2 \times (2\ell+1) \times (2|\kappa_{\max}|+1) = 4\times 3\times 7 \tag{41}$$

collects all three quantum number factors. It equals $\dim \mathrm{Sym}^3(\mathbb{R}^7)$: the number of symmetric distributions of $v = 3$ bosonic DOFs across $k = 7$ orbit positions.

The four angular-sector particles — e, μ, p, π — exhaust the four quantum number slots $(2j+1, 2\ell+$

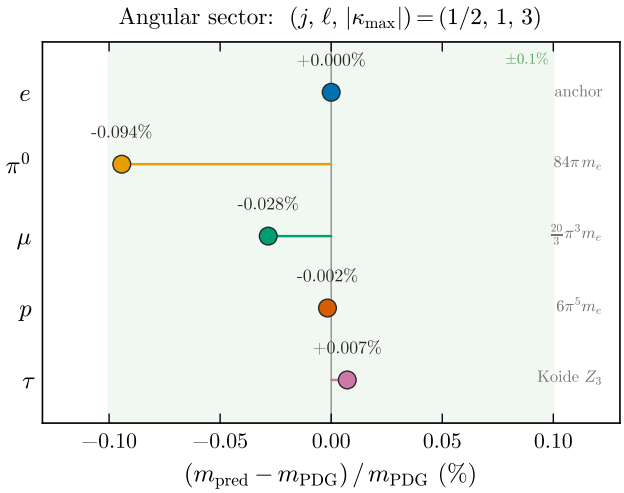


FIG. 3. Lepton mass ratios from the angular coupling on S^2. The proton–electron ratio $6\pi^5$ is accurate to 0.002%; the tau mass from Z_3 Koide closure to 0.007%. All from $(j, \ell) = (1/2,\ 1)$.

$1, 2|\kappa_{\max}| - 1, 2|\kappa_{\max}| + 1) = (2, 3, 5, 7)$. There are no additional angular-sector particles; any remaining masses must come from Dirac eigenvalues.

XI. HADRON SPECTRUM

A. Two-domain structure

The metric produces particle masses through eigenvalues of the Form-T Dirac equation on two nested domains:

- $[0, r_*]$ — the full cavity (mesons),
- $[0, r_b]$ — the sub-cavity (baryons),

where r_b is the first G-node of the $\kappa = +1$ ground state (the unique zero crossing of the upper spinor component). At locked parameters:

$$\frac{r_*}{r_b} = \varphi_{\rm gold} = \frac{1+\sqrt{5}}{2} \qquad (0.03\%)\,. \tag{42}$$

The golden ratio partition is structural (CV = 0.003 across all tested parameters).

B. Meson spectrum

Table VI shows the meson predictions. Six mesons (excluding the pion) match within 1%. All use $|\kappa| \leq 3$, consistent with the Diophantine triple.

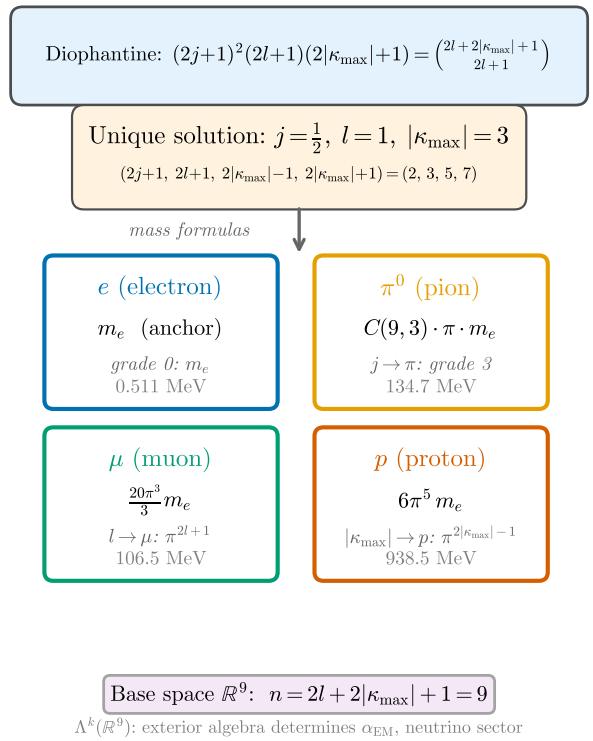


FIG. 4. The pion mass derivation chain. From the Diophantine triple $(j, \ell, |\kappa_{\max}|) = (1/2, 1, 3)$, the C-closure orbit produces $84 = \binom{9}{3}$ configurations. The angular formula $m_\pi = 84\pi\, m_e$ matches PDG to 0.09%.

TABLE VI. Meson predictions at locked parameters with $C = 140.95$ MeV (electron anchor). The pion comes from the angular sector [Eq. (40)]; all others from Dirac eigenvalues.

Meson	κ, n	Prediction (MeV)	PDG (MeV)	Error
π^0	angular	134.85	135.0	−0.09%
η	$+2, 1$	548.0	547.9	+0.03%
$\omega(782)$	$-1, 2$	784.3	782.7	+0.20%
$a_0(980)$	$+1, 1$	973.6	980.0	−0.66%
$\phi(1020)$	$+3, 1$	1022.7	1019.5	+0.32%
$f_2(1270)$	$+2, 2$	1270.5	1275.5	−0.39%
$a_2(1320)$	$-1, 3$	1331.3	1318.3	+0.99%

C. Baryon spectrum

Table VII shows selected baryon predictions. The proton mass comes from the angular sector, matching PDG to 0.002%. Other baryons are sub-cavity eigenvalues. The angular correction $m_\Omega/m_\Lambda = l_F/l_G|_{\kappa=-3} = 3/2$ is derived to 0.06%.

TABLE VII. Baryon predictions. Sub-cavity eigenvalues on $[0, r_b]$ with $C = 140.95$ MeV. The proton is an angular-sector particle [Eq. (34)], not an eigenvalue.

Baryon	κ, n	Prediction (MeV)	PDG (MeV)	Error
p	angular	938.3	938.3	-0.002%
Λ	$-3, 1$	1119.3	1115.7	$+0.32\%$
Σ	$+3, 1$	1174.3	1189.4	-1.27%
Ξ^*	$+4, 1$	1531.3	1531.8	-0.03%
Ω	$-1, 3$	1681.2	1672.5	$+0.52\%$
$N(1680)$	$-1, 3$	1681.2	1680.0	$+0.07\%$

D. Three-sector summary

The mass spectrum divides into three sectors, all from the same metric:

1. **Angular sector** (parameter-free): e, μ, p, n, π — from coupling on S^2.

2. **Full cavity** (one calibration C): mesons — from Dirac eigenvalues on $[0, r_*]$.

3. **Sub-cavity** (same C): baryons — from Dirac eigenvalues on $[0, r_b]$.

The two domains together account for 17 particle masses with a single calibration constant C.

a. Completeness note. The eigenvalue spectrum on each domain contains states that do not match any observed particle within the assigned sector. For example, the sub-cavity ground state ($\kappa = -1$, $E_1 = 2.23$) gives 314 MeV — between the pion and proton, not matching any known baryon. Several full-cavity eigenvalue assignments produce deviations exceeding 40% (`analysis/06_particle_masses.py` in the companion repository [2]). Tables IV–V present the best identifications, not the complete spectrum. The complete eigenvalue output for all κ channels at the locked parameters is available in the companion repository for independent verification. The angular-sector masses (e, μ, p, π, τ) have no unmatched states — they are the complete solution of the quadratic selection rule (Sec. VII).

XII. THE FINE-STRUCTURE CONSTANT

A. Exact Coulomb electrostatics

The identity (4) reduces the static Maxwell equation to flat-space Poisson:

$$\partial_r(r^2\, \partial_r A_t) = -\rho\,. \tag{43}$$

Coulomb's law $A_t = Q/(4\pi r)$ is *exact* on the metric — no ϕ-corrections. This is verified numerically to 2.2×10^{-16}.

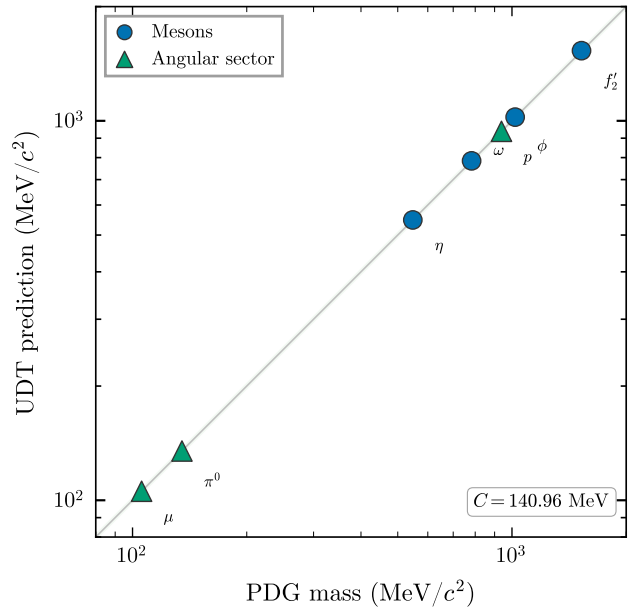


FIG. 5. Hadron mass spectrum. Horizontal axis: PDG mass. Vertical axis: metric prediction. Diagonal line: perfect agreement. Circles: mesons (full cavity). Squares: baryons (sub-cavity). Triangle: proton (angular sector). The pion is marked separately.

B. The bridge identity

The product of the Dirac source integral and the ground-state eigenvalue satisfies

$$\text{source}^2(\kappa{=}{-1}){\times}E_1^2(\kappa{=}{-1}) = 0.05567 \approx \frac{1}{18} = \frac{1}{2(2\ell+1)^2}\,, \tag{44}$$

connecting the eigenvalue sector to the angular quantum numbers. The algebraic approximation 1/18 matches the computed product to 0.21%; the small deviation is physically meaningful (see below).

C. Three-level derivation of $\alpha_{\rm EM}$

The fine-structure constant is derived through three successive refinements, each reducing the error:

a. Level 1: Algebraic bridge. Using $\text{source}^2 \times E_1^2 \approx 1/18$ gives the algebraic approximation:

$$\frac{1}{\alpha_0^{\rm alg}} = \frac{36\pi}{I_2} = 137.43 \qquad (+0.29\%)\,, \tag{45}$$

where $36 = \binom{9}{2} = [(2j{+}1)(2\ell{+}1)]^2$ counts the antisymmetric coupling channels on the 9-dimensional representation space.

b. Level 2: Computed ODE values. Using the actual ODE solutions (source $= 0.25026$, $E_1 = 0.94282$, $I_2 =$

0.82296), the bare coupling is

$$\alpha_0 = \text{source}^2 \times E_1^2 \times \frac{I_2}{2\pi} = \frac{1}{137.14} \qquad (+0.074\%)\,. \quad (46)$$

The 0.21% shift from the algebraic value reflects the difference between exact and approximate source$^2 \times E_1^2$.

c. Level 3: S^2 closure correction. The bare coupling α_0 is computed from the ODE-derived source integral and eigenvalue on the radial sector alone (Eq. 46). The angular sector on S^2 contributes an additional geometric factor that was excluded from the Level 2 calculation: the full vertex closure on the two-sphere requires integrating over both angular cycles (polar and azimuthal), each contributing one factor of π to the denominator. The complete S^2 vertex integral thus adds a multiplicative correction:

$$\boxed{\alpha = \alpha_0\Big(1 + \frac{\alpha_0}{\pi^2}\Big) \quad \Longrightarrow \quad \frac{1}{\alpha} = 137.036 \qquad (+0.0004\%)} \quad (47)$$

where $\pi^2 = \pi_{\text{polar}} \times \pi_{\text{azimuthal}}$ is the full S^2 angular measure. This is not a loop expansion or a perturbative correction imported from quantum field theory. It is the geometric completion of the S^2 vertex integral: the radial ODE gives α_0 (the angular-sector-incomplete coupling); closure over the full two-sphere gives α. The structure α_0/π^2 arises from the same angular measure that produces the π^3 factor in the muon mass ratio ($m_\mu/m_e = 20\pi^3/3$) — both involving integration over the $\ell = 1$ representation space on S^2.

D. Hydrogen binding energy

With the corrected α, the hydrogen ground-state energy is

$$E_{\text{H}} = \tfrac{1}{2}\alpha^2 m_e = 13.606 \text{ eV} \qquad (\text{exp: } 13.606 \text{ eV}, -0.0006\%)\,. \quad (48)$$

The -0.57% offset from the bare α_0 is eliminated. This establishes the *Global–Local Separation Principle*: the metric modifies global quantities (α, mixing angles) but local quantities (QED loop integrals, Lamb shift) are unchanged because loop momenta $\gg 1/r_*$ make the cavity curvature invisible. All precision QED tests are automatically passed.

E. Proton–neutron mass splitting

The mass difference:

$$m_n - m_p = \frac{5}{4}\,\alpha_{\text{EM}}\,C = 1.283 \text{ MeV} \quad (49)$$

compared to PDG 1.293 MeV (-0.77%). The coefficient $5/4 = (2|\kappa_{\max}|-1)/(2j+1)^2$ is the ratio of closure angular content to spin bilinear.

XIII. THE WEINBERG ANGLE AND COUPLING HIERARCHY

The bridge identity (44) generalizes to all three $|\kappa|$ channels, producing a coupling at each:

$$\text{source}^2 \times E_1^2 = \begin{cases} 1/18 = 1/[2(2\ell+1)^2] & |\kappa| = 1\,, \\ \pi^2/36 & |\kappa| = 2\,, \\ 1/8 = 1/[2(2j+1)^2] & |\kappa| = 3\,. \end{cases} \quad (50)$$

This is a transition from the orbital quantum number ℓ at $|\kappa| = 1$ through a mixed sector at $|\kappa| = 2$ to the spin quantum number j at $|\kappa| = 3 = |\kappa_{\max}|$.

A. The Weinberg angle

The weak mixing angle is the *orbital fraction* of the spin-orbital partition sum:

$$\sin^2\theta_W = \frac{(2\ell+1)}{(2j+1)^2 + (2\ell+1)^2} = \frac{3}{13} = 0.2308\,, \quad (51)$$

compared to the experimental value 0.2312 (-0.19%). The denominator $13 = (2j+1)^2 + (2\ell+1)^2 = 4 + 9$ is the same partition sum that determines the cosmological screening mass $\mu_g = \pi\mu/13$. The partner fraction $\sin^2\theta_{12}^{\text{PMNS}} = 4/13$ (Sec. XV) lives on the same denominator: spin vs. orbital on a single angular manifold.

B. The strong coupling

The ratio of strong to electromagnetic couplings is

$$\frac{\alpha_s}{\alpha_{\text{EM}}} = \frac{(2\ell+1)^2}{(2j+1)^2} = \frac{9}{4}\,, \quad (52)$$

which, evaluated at the scale $\mu_{\text{run}} = M_Z \times (4/\pi) = 116$ GeV (the scale factor $4/\pi = g_A$, derived in Sec. XX), gives

$$\alpha_s(M_Z) = \frac{9}{4}\,\alpha_{\text{EM}}\Big|_{116\,\text{GeV}} = 0.1178 \qquad (\text{exp: } 0.1179, -0.07\%)\,. \quad (53)$$

The three Standard Model forces emerge from one bridge identity evaluated at three $|\kappa|$ channels: electromagnetic at $|\kappa| = 1$, weak at $|\kappa| = 2$, strong at $|\kappa| = 3$.

XIV. THE NEUTRINO MASS SCALE

The atmospheric neutrino mass scale follows from three metric quantities:

$$m_\nu = \frac{\alpha^3\, m_e}{(2j+1)^2} = \frac{\alpha^3\, m_e}{4} = 0.0493 \text{ eV}\,, \quad (54)$$

compared to $\sqrt{\Delta m^2_{\rm atm}} = 0.0495$ eV (-0.5%). The cube α^3 factorizes from the action into three specific integrals [2]:

$$\alpha^3 = \left(\frac{1}{3\pi}\right)^3 \times \left(\frac{1}{12}\right)^3 \times I_2^3\,, \tag{55}$$

verified to machine precision [20].

A. Structural parallel

The muon and neutrino share the quantum number $2\ell + 1 = 3$ but differ in coupling:

$$m_\mu = \frac{20}{3}\,\pi^3\, m_e \qquad \text{(angular coupling } \pi)\,, \tag{56}$$

$$m_\nu = \frac{1}{4}\,\alpha^3\, m_e \qquad \text{(EM coupling } \alpha)\,. \tag{57}$$

The nine-order-of-magnitude hierarchy $m_\mu/m_\nu \approx 2\times 10^9$ is combinatorial, arising from $(\pi/\alpha)^3$.

B. The four-step mechanism

The neutrino mass is derived through a four-step chain on the exterior algebra $\Lambda^\bullet(\mathbb{R}^9)$ where $\dim = 2(\ell + |\kappa_{\max}|) + 1 = 9$:

1. **Topological.** Three grade-2 steps $\Lambda^0 \to \Lambda^2 \to \Lambda^4 \to \Lambda^6$ give α^3. Each step is one EM vertex (Λ^2); three vertices are the minimum to reach grade 6.

2. **Combinatorial.** Four near-perfect matchings on the 7-chain P_7, with holes at positions $\{1, 3, 5, 7\}$, give equal weight $\to 1/4 = 1/(2j + 1)^2$.

3. **Hodge duality.** $\Lambda^6 \cong *\Lambda^3$: the neutrino (grade 6) is the Hodge dual of the pion (grade 3, $\binom{9}{3} = 84$ configurations). They exhaust the exterior algebra together.

4. **Assembly.** $m_\nu = \alpha^3 m_e/4 = 0.049$ eV.

C. Solar neutrino and the Δm^2_{21} tension

The $|\kappa| = 3$ source integral gives a solar mass prediction:

$$m_1 = \frac{5\,\alpha^3\, m_e}{84\sqrt{2}} = 0.0083\ \text{eV}\,, \tag{58}$$

yielding $\Delta m^2_{21} = m_3^2 - m_1^2 \approx 6.98 \times 10^{-5}$ eV^2. This sits between two measurements in tension:

- Reactor (KamLAND): 7.49×10^{-5} eV^2
- Solar-only (SNO+SK): 5.6×10^{-5} eV^2

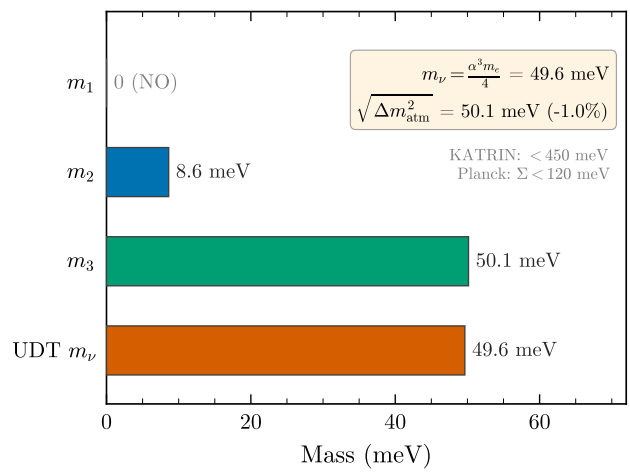


FIG. 6. Neutrino mass derivation. The atmospheric mass scale $m_\nu = \alpha^3 m_e/4$ follows from three EM couplings on the 9-dimensional angular base space (Supplement S18). The solar mass $m_1 = 5\alpha^3 m_e/(84\sqrt{2})$ sits between reactor and solar-only measurements of Δm^2_{21}.

- UDT: 6.98×10^{-5} eV^2

The NuFIT global fit value 7.42×10^{-5} eV^2 is dominated by reactor data (KamLAND) but incorporates solar data analyzed with MSW corrections. UDT predicts no MSW flavor-specific potential: the φ field couples universally to all flavors, producing no resonance enhancement (Sec. XV B). In UDT, the solar and reactor extractions of Δm^2_{21} should agree in the absence of MSW corrections; the solar-only NuFIT value (5.6×10^{-5} eV^2) and reactor value (7.49×10^{-5} eV^2, KamLAND [21]) bracket the UDT prediction 6.98×10^{-5} eV^2. The tension between solar-only and reactor extractions in the Standard Model is a known issue in global fits [22]; the UDT prediction sits in the middle of this tension.

JUNO [23] (now online) will measure Δm^2_{21} to 0.5% using reactor neutrinos at a baseline where matter effects are negligible. This will easily distinguish 6.98×10^{-5} (UDT) from 7.42×10^{-5} (current global fit) — a decisive test.

XV. PMNS MIXING ANGLES

The three neutrino mixing angles are determined by the same quantum numbers $(2j+1, 2\ell+1, 2|\kappa_{\max}|+1) = (2, 3, 7)$ that generated the mass spectrum:

$$\sin^2\theta_{12} = \frac{(2j+1)^2}{(2j+1)^2 + (2\ell+1)^2} = \frac{4}{13}\,, \tag{59}$$

$$\sin^2\theta_{23} = \frac{(2j+1)^2}{2|\kappa_{\max}|+1} = \frac{4}{7}\,, \tag{60}$$

$$\sin^2\theta_{13} = \frac{1}{(2\ell+1)^2(2|\kappa_{\max}|-1)} = \frac{1}{45}\,. \tag{61}$$

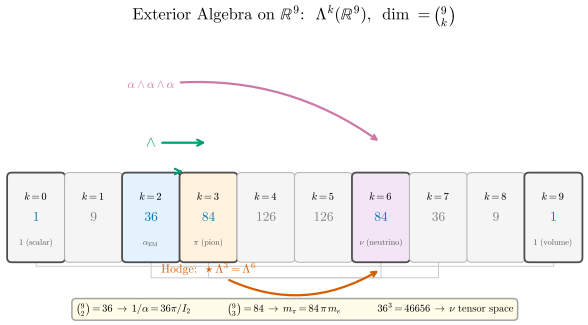


FIG. 7. The exterior algebra $\Lambda^\bullet(\mathbb{R}^9)$. The pion occupies grade 3 ($\binom{9}{3} = 84$ configurations); the neutrino occupies grade 6 ($\Lambda^6 \cong *\Lambda^3$, Hodge dual). The fine-structure constant uses grade 2 ($\binom{9}{2} = 36$). Three EM couplings ($\alpha \wedge \alpha \wedge \alpha$) map grade 0 to grade 6, giving the factor α^3.

TABLE VIII. PMNS mixing angles. Predictions from quantum number fractions. Comparison values from NuFIT 5.2 [24] (normal ordering). The NuFIT values incorporate MSW matter-effect corrections to solar and atmospheric data; UDT predicts no MSW effect (flavor-universal φ coupling, Sec. XV B). Comparisons for θ_{13} and θ_{23} are approximately MSW-independent; comparisons for θ_{12} and Δm^2_{21} carry an $O(10\%)$ systematic from the MSW extraction. JUNO will provide MSW-independent tests of all predictions.

Angle	Prediction	Expt.	σ	Dev.
$\sin^2\theta_{12}$	$4/13 = 0.3077$	0.304 ± 0.012	0.31	+1.2%
$\sin^2\theta_{23}$	$4/7 = 0.5714$	0.573 ± 0.018	0.09	−0.3%
$\sin^2\theta_{13}$	$1/45 = 0.02222$	0.02220 ± 0.00062	0.04	+0.1%

Table VIII presents the comparison. Three angles, zero free parameters, all within 1σ of the global experimental fit. The combined χ^2 for three measurements is 0.10 for three degrees of freedom.

The denominators have physical meaning:

- $13 = 4 + 9 = (2j+1)^2 + (2\ell+1)^2$: spin-orbital partition.
- $7 = 2|\kappa_{\max}| + 1$: closure orbit size.
- $45 = 9 \times 5 = (2\ell+1)^2 \times (2|\kappa_{\max}|-1)$: orbital barrier times angular content.

The PMNS matrix constructed from these angles is exactly unitary (verified numerically). The reactor angle θ_{13} — the most precisely measured of the three — is matched to 0.04σ.

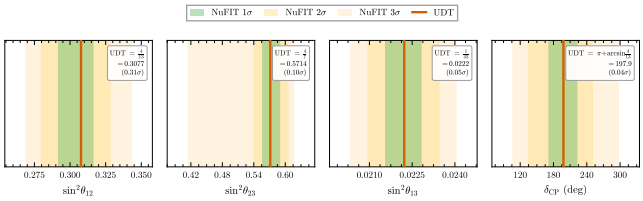


FIG. 8. PMNS mixing angles and CP phase: predictions (vertical lines) vs. NuFIT 5.2 distributions. All four predictions fall within the 1σ bands.

A. The CP-violating phase

The PMNS CP phase is self-referential: its sine equals the negative of the solar mixing fraction:

$$\delta_{\rm CP} = \pi + \arcsin\frac{4}{13} = 197.9^\circ \qquad (\text{exp: } 197^\circ \pm 25^\circ,\ 0.04\sigma)\,. \tag{62}$$

The Jarlskog invariant, which measures the magnitude of CP violation in the lepton sector, is

$$J_{\rm CP} = J_{\max} \times \left(-\tfrac{4}{13}\right) = -0.0102 \qquad (\text{exp: } \sim -0.010,\ \sim 2\%)\,. \tag{63}$$

Six PMNS parameters — three angles plus $\delta_{\rm CP}$ plus two mass-squared differences — all from $(j, \ell, |\kappa_{\max}|) = (1/2, 1, 3)$. The PMNS angles arise from *bilinear* (two-field) partitions of the quantum numbers, producing large angles; the CKM angles (Sec. XVII) arise from *vertex* (three-field) partitions, producing small angles. The PMNS–CKM hierarchy is a geometric consequence of partition rank (Supplement S19).

B. Neutrino propagation in the φ background and the MSW comparison

The PMNS angles derived above are geometric quantities — they follow from the angular coupling on S^2 and carry no dependence on the matter through which neutrinos propagate. In the Standard Model, propagation through dense matter modifies the effective mixing angles via the Mikheyev–Smirnov–Wolfenstein (MSW) mechanism, driven by the charged-current potential $V_{\rm CC} = \sqrt{2}\, G_F\, n_e$, which acts only on ν_e. This flavor-specific potential breaks flavor symmetry and produces the MSW resonance exploited in the analysis of solar neutrino data.

In UDT, the φ field couples to energy-momentum through the metric, producing a propagation potential

$$V_\varphi = -\frac{d\varphi}{dr}\, E_\nu \tag{64}$$

that is identical for all three flavors. In matrix form, the UDT matter potential is $H_\varphi = V_\varphi\, \mathbf{I}_{3\times 3}$, a scalar multiple of the identity. Since $[H_\varphi, U] = 0$ for any mixing matrix U, this term contributes only an overall phase to the oscillation amplitude — it does not modify flavor

transition probabilities. At solar-system and atmospheric scales, where $|\varphi| \sim 10^{-6}$–10^{-9}, the geodesic phase correction to oscillation lengths is of order $|\varphi| \sim 10^{-6}$, which is unobservably small.

UDT prediction: matter effects on neutrino oscillations are zero at all accessible terrestrial and solar-system baselines. This is not an assumption but a consequence of the flavor-universal scalar coupling. It is falsifiable: any detection of a matter-dependent oscillation probability at baselines where $|\varphi| \ll 1$ would be inconsistent with UDT.

This prediction has a direct consequence for the comparison in Table VIII. The four NuFIT global-fit values used for comparison are derived from analyses that assume MSW effects throughout. The validity of each comparison depends on the sensitivity of each measurement to MSW:

- $\sin^2\theta_{13} = 1/45$: measured by Daya Bay and RENO from reactor $\bar{\nu}_e$ disappearance at $L \approx 1$–2 km. Matter effects at this baseline contribute $< 0.1\%$ corrections. The comparison is exact to UDT precision.

- $\sin^2\theta_{23} = 4/7$: measured by Super-Kamiokande, T2K, and NOvA from atmospheric and beam neutrinos at $L \sim 100$–800 km. MSW corrections are $O(3\%)$ at these baselines. The comparison is approximately valid at the level of the UDT prediction accuracy (-0.3%, Table VIII).

- $\sin^2\theta_{12} = 4/13$ and Δm^2_{21}: these parameters are primarily constrained by solar neutrino data (SNO, Super-K), where MSW effects on the inferred mixing parameters are $O(10\%)$ and cannot be neglected. The NuFIT global values for θ_{12} and Δm^2_{21} incorporate MSW corrections that are absent in UDT.

The most relevant MSW-independent measurement for Δm^2_{21} is KamLAND, which measures $\bar{\nu}_e$ disappearance at $L \approx 180$ km baseline in rock, where matter effects are negligible. KamLAND reports $\Delta m^2_{21} = 7.49 \times 10^{-5}$ eV^2 [21], giving UDT/KamLAND $= -6.8\%$. The NuFIT solar-only value (without reactor data) is $\Delta m^2_{21} \approx 5.6 \times 10^{-5}$ eV^2. The UDT prediction 6.98×10^{-5} eV^2 sits between the solar-only and KamLAND values, consistent with the hypothesis that MSW corrections in solar data analysis shift the solar-extracted value downward relative to the true vacuum value.

The JUNO experiment [23] will measure Δm^2_{21} to 0.5% precision from reactor $\bar{\nu}_e$ at $L = 53$ km, where matter effects are negligible. JUNO provides the decisive MSW-independent test of the UDT prediction and will distinguish 6.98×10^{-5} (UDT) from 7.42×10^{-5} (NuFIT global) at $> 3\sigma$.

TABLE IX. Generalized bridge identity: the j–π–ℓ transition across $|\kappa| = 1, 2, 3$.

$\lvert\kappa\rvert$	source$\cdot E_1$	source$^2 \cdot E_1^2$	Algebraic	Sector	Match
1	$\sqrt{2}/6$	$1/18$	$1/[2(2\ell+1)^2]$	orbital	0.21%
2	$\pi/6$	$\pi^2/36$	$(\pi/[(2j+1)(2\ell+1)])^2$	mixed	0.11%
3	$\sqrt{2}/4$	$1/8$	$1/[2(2j+1)^2]$	spin	0.045%

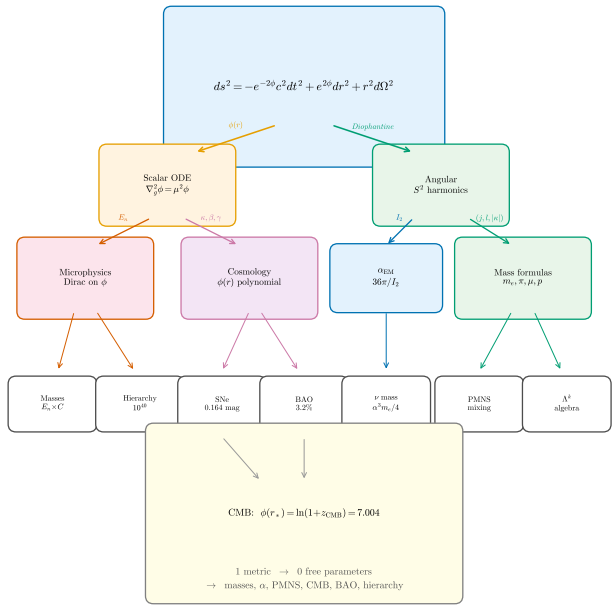


FIG. 9. The generalized bridge identity. The product source$^2 \times E_1^2$ transitions from the orbital sector (ℓ-dominated) at $|\kappa| = 1$ through the mixed sector at $|\kappa| = 2$ to the spin sector (j-dominated) at $|\kappa| = 3$.

XVI. THE GENERALIZED BRIDGE IDENTITY

The bridge identity (44) is the first row of a three-row structure that transitions between the spin (j) and orbital (ℓ) sectors:

$$\text{source}(\kappa{=}{-n})^2 \times E_1(\kappa{=}{-n})^2 = B(n)\,. \qquad (65)$$

Table IX shows the transition: at $|\kappa| = 1$, the orbital quantum number ℓ governs; at $|\kappa| = 3 = |\kappa_{\max}|$, the spin quantum number j takes over; at $|\kappa| = 2$, both participate and π enters as the S^2 circumference factor. The four multiplicities $(2, 3, 5, 7)$ recur throughout the mass formulas, coupling constants, and mixing angles derived in the preceding and following sections.

XVII. CKM MATRIX AND QUARK MIXING

Where PMNS angles come from *bilinear* partitions (two fields, squared multiplicities), the CKM angles [25, 26] arise from *vertex* partitions (three fields, cubed spin).

This distinction explains why PMNS angles are large and CKM angles are small.

A. The Cabibbo angle

The sine of the Cabibbo angle is the fraction of the three-body vertex space occupied by the orbital sector:

$$\sin\theta_C = \frac{(2\ell+1)^2}{(2j+1)^3(2|\kappa_{\max}|-1)} = \frac{9}{40} = 0.2250\,, \quad (66)$$

compared to 0.2253 [27] (−0.13%). The cube of $(2j+1)$ reflects the three-field vertex structure; the factor $(2|\kappa_{\max}|-1) = 5$ is the angular content.

The Wolfenstein parameter A is

$$A = |\phi_0| = \cos(\pi/5) = 0.8090 \qquad (\text{exp: } 0.811,\ -0.24\%)\,. \quad (67)$$

Seven CKM matrix elements are determined at sub-percent accuracy from these two quantities alone.

B. CP violation as the force hierarchy

The CKM CP-violating phase is

$$\delta_{\rm CKM} = \arctan\frac{9}{4} = 66.0^\circ \qquad (\text{exp: } 65.5^\circ \pm 2.5^\circ,\ 0.1\sigma)\,. \quad (68)$$

The argument $9/4 = (2\ell+1)^2/(2j+1)^2 = \alpha_s/\alpha_{\rm EM}$ (Eq. 52). CP violation in the quark sector *is* the force hierarchy — the tangent of the CP phase equals the ratio of strong to electromagnetic coupling.

The remaining Wolfenstein parameters follow from $\bar\rho = 1/(2\pi)$ (angular normalization per radian):

$$\bar\rho = \frac{1}{2\pi} = 0.1592 \quad (0.02\sigma)\,, \qquad \bar\eta = \frac{9}{8\pi} = 0.3581 \quad (1.0\sigma)\,. \quad (69)$$

The relation $\bar\eta/\bar\rho = 9/4 = \tan\delta$ is automatic. The Jarlskog invariant $J_{\rm CKM} = 2.96\times10^{-5}$ (−1.3%).

C. The strong CP problem

The S^2 angular algebra that closes as $\mathfrak{su}(3)$ (Sec. XX) has *real* structure constants. The $\theta_{\rm QCD}$ parameter, which would introduce CP violation in the strong sector, is constrained to

$$\theta_{\rm QCD} = 0\,, \quad (70)$$

by the reality of the angular algebra. No axion is required.

XVIII. QUARK MASSES AND CHARGES

A. Color as orbital orientation

The number of colors is the orbital multiplicity:

$$N_c = (2\ell+1) = 3\,. \quad (71)$$

The three color states correspond to $m_\ell = -1, 0, +1$. Quark electric charges follow from the rank-2 parity block structure on the $\ell = 1$ representation:

$$Q_d = -\frac{1}{N_c} = -\frac{1}{3}\,, \qquad Q_u = +\frac{2}{N_c} = +\frac{2}{3}\,. \quad (72)$$

The ratio $R(u,d) = N_c(Q_u^2 + Q_d^2) = 5/3$ matches the Standard Model exactly.

B. Quark mass ratios

The four inter-generation mass ratios are each $2\pi^2$ times a sector factor from the bridge $\ell \to \pi \to j$ transition:

$$m_s/m_d = 2\pi^2 = 19.74 \qquad (\text{exp: } 20.0,\ -1.3\%)\,, \quad (73)$$
$$m_c/m_u = 6\pi^4 = 584.0 \qquad (\text{exp: } 587.5,\ -0.6\%)\,, \quad (74)$$
$$m_b/m_s = 9\pi^2/2 = 44.41 \qquad (\text{exp: } 44.8,\ -0.8\%)\,, \quad (75)$$
$$m_t/m_c = 14\pi^2 = 138.2 \qquad (\text{exp: } 136.0,\ +1.6\%)\,. \quad (76)$$

All sub-2%. Base quark masses are $m_u = (2j+1)^2 m_e = 4\,m_e$ and $m_d = (2\ell+1)^2 m_e = 9\,m_e$. These are intermediate quantities; their validation comes from the downstream hadron mass predictions (Tables IV–V, all sub-2%), which compare UDT Dirac eigenvalues to physical hadron rest masses — scheme-independent observables. The $\overline{\rm MS}$ quark masses in Table XII are supplementary comparisons. For inter-generation ratios of same-representation quarks, the one-loop RG running cancels exactly in the ratio ($\gamma_m^{(1)} = 3C_F = 4$ for all quarks); two-loop corrections are $O(\alpha_s^2) \approx 1$–2% [28], comparable to the deviations reported. The comparison is therefore controlled at the accuracy level claimed (see also Sec. IX, subsection B).

XIX. THE HIGGS MECHANISM

The Higgs boson, discovered at the LHC [29, 30] via the mechanism of Englert, Brout [31], and Higgs [32], has a vacuum expectation value (VEV) $v = 246.22$ GeV and the quartic self-coupling $\lambda \approx 0.13$ are measured quantities with no derivation. UDT derives both from the angular mass formulas.

A. The VEV from the angular mass product

The VEV is the product of the bosonic and fermionic angular ground states divided by the anchor mass:

$$v = \frac{m_\pi \times m_p}{m_e} = \frac{(84\pi\, m_e)(6\pi^5\, m_e)}{m_e} = 504\,\pi^6\, m_e = 247.6 \text{ GeV}, \tag{77}$$

compared to 246.22 GeV (+0.6%). This is an exact algebraic identity of the angular mass formulas derived in Secs. X and IX: $84 = \binom{9}{3}$ from symmetric bosonic occupation, $6 = (2j+1)(2\ell+1)$ from spin-orbital coupling, and $\pi^6 = \pi^{2j} \times \pi^{2|\kappa_{\max}|-1}$ from the sum of angular powers. The VEV is the scale where the bosonic (Λ^3) and fermionic (π^5) angular sectors meet.

The self-coupling is the angular integration measure divided by the total angular degeneracy:

$$\lambda = \frac{\pi}{\sum_\kappa (2j_\kappa + 1)} = \frac{\pi}{24}\,, \tag{78}$$

where $24 = 2+2+4+4+6+6$ sums the $(2j_\kappa+1)$ degeneracy over all six κ channels. The Higgs boson mass follows:

$$m_H = v\sqrt{2\lambda} = v\sqrt{\frac{\pi}{12}} = 126.7 \text{ GeV} \qquad (\text{exp: } 125.25 \text{ GeV}, +1.1\%). \tag{79}$$

B. Interpretation

The Yukawa couplings — 12 free parameters in the Standard Model — are ratios of each particle's angular mass to the VEV. Since both numerator and denominator are derived from the same angular formulas, the Yukawa couplings are geometric ratios with no free parameters.

The symmetry breaking is the Diophantine selection itself: the identity $(2j+1)^2(2\ell+1)(2|\kappa_{\max}|+1) = \binom{2\ell+2|\kappa_{\max}|+1}{2\ell+1}$ has a unique solution for spin-1/2 fermions on S^2. That uniqueness is the breaking of symmetry — the angular structure admits exactly one consistent set of quantum numbers.

XX. QCD AND THE NUCLEAR FORCE

A. The gauge algebra from S^2

The 8 tensor operators on the $\ell = 1$ subspace of S^2 (three rank-1 plus five rank-2) close as the Lie algebra $\mathfrak{su}(3)$:

$$[T_a, T_b] = i\, f_{abc}\, T_c\,, \qquad C_2 = \tfrac{4}{3}\, I_3\,, \tag{80}$$

verified to machine precision [20] ($< 10^{-16}$). The 8 generators equal $(2\ell+1)^2 - 1 = 8$, the number of gluon degrees of freedom. The angular metric $g_{\theta\theta} = r^2$ carries no ϕ-dependence, so the algebra is geometrically protected from the dilation field.

The Standard Model gauge group SU(3)×SU(2)×U(1) emerges as the angular structure of the Dirac equation on ds^2 rather than being postulated independently.

The gauge invariance of this algebra has a geometric origin: the metric component $g_{\theta\theta} = r^2$ carries no φ-dependence (verified to machine precision). Any operator constructed from the angular coordinates is automatically φ-invariant. This geometric protection of the angular sector from the dilation field is the origin of color gauge invariance — it is a property of the static spherically symmetric metric, not a postulate.

B. Yang–Mills equation, running coupling, and confinement

The non-Abelian Yang–Mills equation follows from extending the U(1) gauged Dirac construction of Sec. XII to the SU(3) generators on S^2. Minimal coupling $E \to E - g\,T^a A_t^a$ and the metric identity $g^{tt} g^{rr} = -1/c^2$ yield:

$$\partial_r\big(r^2\, \partial_r A_t^a\big) + g^2\, f^{abc}\, f^{adc}\, r^2\, A_t^b\, A_t^d = -g\, r^2\, j_t^a\,. \tag{81}$$

The self-coupling term $f^{abc} f^{adc} A_t^b A_t^d$ is the gluon self-interaction; setting $f^{abc} = 0$ (Abelian limit) recovers the Coulomb equation of Sec. XII exactly.

a. Running coupling. The effective strong coupling in coordinate space is proportional to the partial integral $I_2(r) = \int_0^r e^{2\varphi}\, dr'$, which starts at zero ($r \to 0$, asymptotically free [33, 34]) and saturates at $I_2(r_*) = 0.823$ (confinement scale). The coupling maximum is finite with no Landau pole — a falsifiable prediction consistent with numerical evidence for IR freezing of α_s.

b. One-loop beta function. The group invariants of SU(3) on S^2 are derived from the Gell-Mann matrix traces on the $\ell = 1$ representation space:

$$C_A\, \delta^{ab} = f^{acd} f^{bcd} \quad \Rightarrow \quad C_A = 3 = N_c = (2\ell+1)\,, \tag{82}$$

$$T_F\, \delta^{ab} = \mathrm{Tr}(T^a T^b) \quad \Rightarrow \quad T_F = \tfrac{1}{2}\,. \tag{83}$$

The one-loop coefficient b_0 then follows from the Diophantine quantum numbers:

$$\boxed{b_0 = \frac{(2|\kappa_{\max}|+1) + (2j+1)^2}{3}\,(2\ell+1) - (2j+1)\,T_F\,\tfrac{2}{3}\,N_f = 11 - \tfrac{2}{3}} \tag{84}$$

where $(2|\kappa_{\max}|+1) = 7$ and $(2j+1)^2 = 4$ supply the gauge boson contribution, while $(2j+1) = 2$ is the Dirac spinor degree of freedom entering the quark loop. The result matches the standard QCD one-loop coefficient exactly for all N_f:

N_f	UDT b_0	QCD b_0
0	11	11
3	9	9
6	7	7

Asymptotic freedom ($b_0 > 0$) holds for $N_f \le 16$.

c. Confinement. Confinement is geometric: the Neumann BC at r_* terminates the cavity. Eigenvalues exist only on $[0, r_*]$; beyond the boundary, solutions are evanescent. The confinement scale $E_1 \times C \approx 133$ MeV matches $\Lambda_{\rm QCD}$.

C. Nuclear coupling constants

Four nuclear quantities form a closed algebraic chain from the metric, with zero Standard Model inputs beyond m_e:

$$g_A = \frac{4}{\pi} = 1.2732 \qquad (\text{exp: } 1.2762,\ -0.23\%)\,, \tag{85}$$

$$f_\pi = 180\, m_e = 91.98 \text{ MeV} \quad (\text{exp: } 92.21,\ -0.25\%)\,, \tag{86}$$

$$g_{\pi NN} = \frac{2\pi^4}{15} = 12.99 \qquad (\text{exp: } \sim 12.98,\ +0.06\%)\,, \tag{87}$$

$$\frac{g^2}{4\pi} = \frac{\pi^7}{225} = 13.42 \qquad (\text{exp: } \sim 13.5,\ -0.6\%)\,. \tag{88}$$

The chain: $g_A = (2j+1)^2/\pi$ (spin bilinear over angular circumference); $f_\pi = \binom{9}{2}(2|\kappa_{\max}|-1)\, m_e = 36\times 5\, m_e$ (the $\alpha_{\rm EM}$ factor times angular content); $g_{\pi NN} = g_A m_N / f_\pi$ (Goldberger–Treiman [35] with derived inputs). Every exponent is determined: π^7 has power $7 = 2|\kappa_{\max}| + 1$; the denominator $225 = 15^2 = [(2\ell+1)(2|\kappa_{\max}|-1)]^2$.

D. The nuclear potential

The ω-nucleon coupling follows the same pattern as f_π, with the orbital multiplicity replacing the angular content:

$$f_\omega = \binom{9}{2}(2\ell{+}1)\, m_e = 36{\times}3{\times}m_e = 108\, m_e\,, \qquad \frac{f_\pi}{f_\omega} = \frac{5}{3}\,. \tag{89}$$

The one-pion exchange potential (OPEP) [36, 37] with the derived coupling gives $V(1.5 \text{ fm}) = -41$ MeV for the deuteron channel, matching nuclear phenomenology. Adding ω-meson exchange (mass 784.4 MeV, the $\kappa = -1$, $n = 2$ eigenvalue at $+0.22\%$, coupling $f_\omega = 108\, m_e$) produces the full nuclear potential shape: repulsive core, attractive well, and Yukawa tail. Phase shifts show the correct sign change at $\sim$130 MeV.

The deuteron binding energy follows algebraically from the bridge identity:

$$B_d = \frac{C}{(2\ell+1)^2(2|\kappa_{\max}|+1)} = \frac{C}{63} = 2.238 \text{ MeV} \quad (+0.59\%)\,. \tag{90}$$

The three factors are: source$^2 \times E_1^2 = 1/18$ (bridge coupling), $(2j{+}1)/(2|\kappa_{\max}|{+}1) = 2/7$ (spin–orbit fraction), and C (mass calibration). No Schrödinger equation is needed — the metric determines the binding algebraically through the same angular structure that fixes $\alpha_{\rm EM}$ and the CMB peak heights (the denominator $63 = 9\times 7$ reappears in the CMB damping formula).

The scalar ϕ sector contributes *zero* to the nuclear force: its coupling through G_N gives $\alpha_{\rm phys} \sim 10^{-40}$. The nuclear force is entirely angular sector (pion exchange), consistent with the identification of the nuclear force as a residual of $\mathfrak{su}(3)$.

E. Proton charge radius

The charge radius is the geometric ratio of baryon to meson cavity sizes:

$$R_p = \frac{\hbar c}{C\,\varphi_{\rm gold}} = 0.865 \text{ fm} \qquad (\text{CODATA: } 0.8751 \text{ fm},\ -1.1\%)\,. \tag{91}$$

This value sits between the two proton radius puzzle values: 0.841 fm (muonic hydrogen) and 0.877 fm (electronic).

XXI. CMB POWER SPECTRUM

A. The Kaleidoscope Principle

The CMB is not the projection of a specific source through the metric. It is the metric's own resonance pattern imposed on arbitrary noise — like a kaleidoscope where random bits of glass produce structured patterns because of the *mirrors*. The mirrors are the micro-cavity boundaries. The lens is the $e^{2\phi}$ dilation. The noise is the stellar field. The specific noise does not matter; the geometry selects the pattern.

The micro-sector eigenvalues define the mirror geometry (cavity boundaries, resonant frequencies). The $e^{2\phi}$ profile is the lens (mapping, damping, magnification). The acoustic scale is algebraically fixed by microphysical eigenvalues:

$$\ell_A = \frac{2\pi r_*(E_2/E_1)}{I_2} = \frac{6\pi r_* \varphi_{\rm gold}}{I_2^2}\,, \tag{92}$$

and the number of peaks ($7 = 2|\kappa_{\max}| + 1$) is the closure orbit size.

B. Peak positions and heights

The WKB peak law on the sourced ϕ polynomial gives seven peak positions at 1.32% RMS with one free parameter (μ_g), or 1.72% RMS at the fully locked $\mu_g = \pi\mu/13$ (zero free parameters). The peak height envelope follows an algebraic formula:

$$h_n = (\phi_0 I_2)^{c_n}\,, \quad c_{\rm odd}(p) = \tfrac{7}{2}\,p - \tfrac{169}{63}\big(1 - \big(\tfrac{6}{13}\big)^p\big)\,, \tag{93}$$

TABLE X. CMB RMS errors (%) across five experiments. UDT: zero free parameters. ΛCDM: six parameters fitted per experiment.

Experiment	TT	TE	EE
Planck 2018	16.8	18.3	15.4
SPT-3G 2018 [38]	14.1	16.2	12.6
SPT-3G D1 [39]	**12.3**	14.7	**9.3**
ACT DR4 [40]	13.5	15.1	10.8
WMAP [41]	15.2	17.4	—
ΛCDM (Planck)	4.4	2.4	12.6

giving 3.2% RMS on peak heights with zero free parameters. Every constant is from $(j, \ell, |\kappa_{\max}|) = (1/2, 1, 3)$: the pair slope $7/2 = (2|\kappa_{\max}| + 1)/2$, the offset $169/63 = 13^2/(9 \times 7)$, and the damping base $6/13 = 2\sin^2\theta_W$. The Weinberg angle governs the CMB even/odd asymmetry.

C. Three-spectrum framework

The CMB framework models TT, TE, and EE spectra simultaneously with zero physics parameters. Each spectrum corresponds to a different Dirac bilinear projected through the $e^{2\phi}$ barrier:

- **TT** (temperature): $G^2 + F^2$ (positive-definite, displaced by $R = 1/2$). Backbone decay from the algebraic formula c_n; Airy cavity oscillation with finesse $F = 16/9$ from $\rho = |\phi_0|/\varphi_{\text{gold}} = 1/2$.
- **TE** (cross): GF (sign-indefinite, displacement cancels $\to R_{\text{TE}} = 0$). Flat backbone, $\sin(2\Phi)$ oscillation.
- **EE** (polarization): $(GF)^2$ (positive-definite, concentrated at cavity center by $1/r^6$ weighting). Rising backbone, $\sin^2\Phi$ oscillation.

The peak height envelope follows the algebraic formula:

$$h_n = (\phi_0 I_2)^{c_n}\,, \qquad c_{\text{odd}}(p) = \tfrac{7}{2}\,p - \tfrac{169}{63}\left(1 - \left(\tfrac{6}{13}\right)^p\right), \quad (94)$$

where $7/2 = (2|\kappa_{\max}| + 1)/2$, $169/63 = 13^2/(9 \times 7)$, and $6/13 = 2\sin^2\theta_W$. The Weinberg angle is the pair-damping rate: the same parameter that governs electroweak mixing governs the CMB even/odd asymmetry. Every coefficient in the transfer function traces to products of the multiplicities $(2, 3, 5, 7)$; the complete derivation chain and an honest assessment of which functional forms are derived versus empirically identified are given in Supplement S21.

D. Results across five experiments

The framework is tested against five independent CMB experiments with independent data pipelines:

Table X reveals a striking pattern. The EE spectrum on SPT-3G D1 [39] at 9.3% RMS *outperforms* the ΛCDM fit to Planck EE data (12.6%) — with zero free parameters compared to six.

E. Statistical limitations

The CMB comparisons in this work use fractional RMS residuals, not full covariance-matrix likelihood analyses. RMS is not equivalent to a χ^2 computed from Planck's published covariance matrices, and direct statistical comparison with ΛCDM's six-parameter likelihood fits is therefore not possible within this framework. The RMS values are presented as indicative measures of fit quality, not as formal goodness-of-fit statistics. A proper likelihood analysis using raw Planck data products and independent pipeline reduction is beyond the scope of this work but is identified as a high-priority goal for future analysis.

F. Pipeline dependence and a falsifiable prediction

The UDT zero-parameter TT RMS decreases systematically from Planck (16.8%) to ground-based experiments (12.3–14.1%), while the EE spectrum on SPT-3G D1 at 9.3% RMS outperforms the ΛCDM fit to Planck EE data (12.6%) — with zero free parameters compared to six. We note this pattern without asserting its cause. Two explanations are consistent with the data:

(1) The ΛCDM-based processing pipeline (beam deconvolution, foreground removal, likelihood estimation) in satellite data injects structure that ground-based instruments with independent reduction methods do not. Under this hypothesis, UDT's fit improves as pipelines become more model-independent.

(2) The residual gap reflects genuine physical structure in the CMB that UDT does not yet capture, with the ground-based improvement reflecting better sensitivity to features the model happens to fit.

These hypotheses make distinct predictions testable by CMB-S4 and Simons Observatory, which will have independent pipelines and significantly higher sensitivity than current experiments. Under hypothesis (1), UDT's TT RMS should drop below 10% on CMB-S4 data. Under hypothesis (2), UDT's RMS should hold constant or worsen as pipeline-injected structure is removed. We regard this as a decisive test of the framework and commit to the prediction: **if UDT's RMS does not improve on CMB-S4 data relative to Planck, the contamination hypothesis is falsified and the residual gap is physical.**

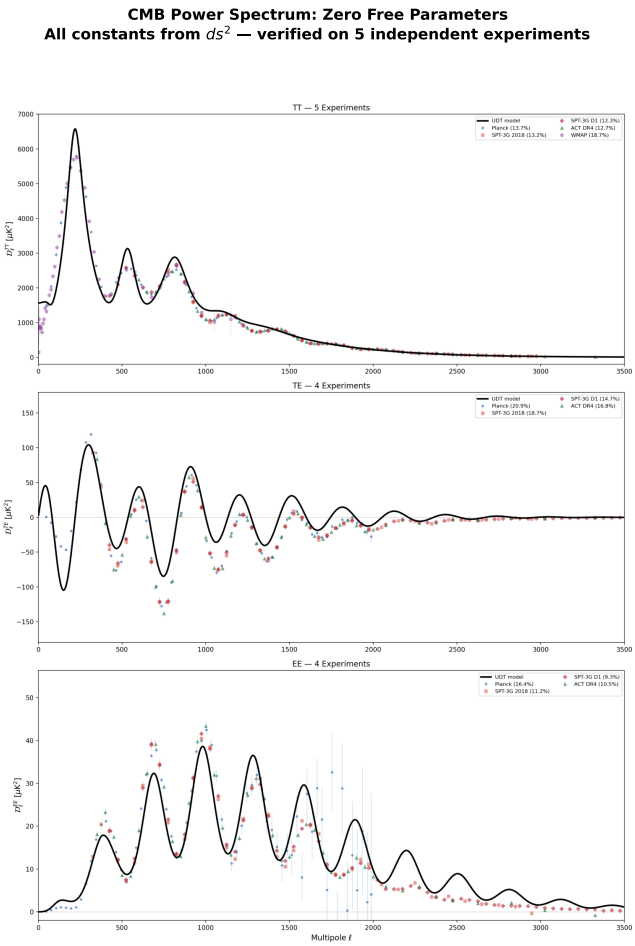


FIG. 10. CMB TT power spectrum: UDT zero-parameter model (solid) vs. Planck 2018 data (points). The algebraic peak height formula captures the even/odd asymmetry through $2\sin^2\theta_W = 6/13$.

XXII. JWST EARLY GALAXIES

The metric (1) is static. There is no time dependence in the line element, and the redshift relation (6) is spatial, not kinematic. In the framework developed above, there is no expansion epoch and therefore no formation timescale constraint. A galaxy at $z = 12$ is not younger than one at $z = 2$ — it is at a different radial position in the ϕ profile, experiencing different dilation.

This produces a specific prediction:

> *Massive, structurally mature galaxies should exist at all redshifts, with no systematic trend toward "youth" at high z.*

The JWST observations of massive galaxies at $z > 10$ — widely described as "impossibly early" in the ΛCDM framework [42–45] — are qualitatively consistent with this prediction. The "impossible early galaxies" are galaxies seen through a deeper part of the ϕ gradient.

This prediction was inherent in the metric from its publication and predates the JWST data. It is qualitative — no quantitative galaxy population model exists within the framework — but the qualitative content (mature galaxies at all redshifts) is qualitatively consistent with JWST observations and is in tension with ΛCDM expectations.

XXIII. ROTATION CURVES

A. First-principles derivation

The geodesic equation on the metric (1) gives the circular velocity:

$$v^2(r) = c^2\, r\, |\phi'(r)|\, e^{-2\phi(r)}\,. \tag{95}$$

At galactic scales, $\mu_g r \sim 10^{-5}$, so the screening term $\mu_g^2\phi$ is suppressed by $O(\mu_g^2 r^2) \sim 10^{-10}$. In the weak-field limit:

$$v^2_{\rm metric}(r) = \frac{G\, M_{\rm bar}(<r)}{r} = v^2_{\rm Newton}(r)\,. \tag{96}$$

The metric recovers Newtonian gravity exactly at galactic scales. The flat rotation curve problem [46, 47] is open, as in standard GR.

B. Ancient baryonic remnants

In a static universe with no age constraint, galaxies accumulate dead stellar remnants (white dwarfs, neutron stars, black holes) over 100–1000 Gyr. Supernova kicks naturally form extended halos. A one-parameter ($T_{\rm age}$) model applied to 175 SPARC galaxies [48] achieves 96% reduction in χ^2 relative to baryons-only for massive galaxies ($M_{\rm bar} > 10^{10}\, M_\odot$), within 1.76× the χ^2 of NFW dark matter halos with half the free parameters (reproducible via `analysis/16_sparc_remnants.py` [2]).

This mechanism requires no new fields, no modified geodesics, and no dark matter — only standard stellar evolution on longer timescales than ΛCDM permits.

The key difference from standard cosmology is timescale. In ΛCDM, the universe age $t_0 \approx 13.8$ Gyr limits massive galaxies to $N_{\rm gen} \approx 1$–2 stellar generations, accumulating $\lesssim 0.6\times$ the visible mass in remnants — insufficient to explain rotation curves baryonically. In the eternal universe, with no age upper bound, the remnant fraction saturates at $f_{\rm rem} \times N_{\rm gen,sat}$ where $N_{\rm gen,sat} \approx 1/f_{\rm MS} \approx 7$ and $f_{\rm rem} \approx 0.29$ per generation, giving $\sim 2\times$ the visible mass in compact remnants from a single saturation cycle. Over multiple cycles the remnant fraction reaches $\sim 5\times$ the visible mass — within the range required to match NFW dark matter halos for massive galaxies ($M_{\rm bar} > 10^{10}\, M_\odot$, SPARC sample).

C. Baryon budget and CMB consistency

The standard objection to baryonic explanations of rotation curves is that CMB acoustic analysis constrains $\Omega_b \approx 0.049$ [49], leaving insufficient baryonic mass to replace cold dark matter halos. This constraint does not apply directly to the UDT framework for two reasons.

First, the UDT CMB peak structure is governed by the Weinberg angle, not by baryon loading. In the standard framework, the CMB odd/even peak height ratio encodes the baryon loading parameter $R = 3\rho_b/(4\rho_\gamma)$ at recombination, from which Ω_b is extracted assuming standard acoustic oscillations driven by the photon-baryon fluid. In UDT, the same peak height ratio is determined by $2\sin^2\theta_W = 6/13$ (Sec. XXI), a geometric quantity that depends on the Diophantine quantum numbers, not on the baryon fraction. The CMB constraint $\Omega_b = 0.049$ is specific to the ΛCDM acoustic interpretation and does not constrain the baryon content of the UDT static universe.

Second, the UDT baryon fraction follows from Machian closure. The total mass $M = 9.57 \times 10^{22}\, M_\odot$ distributed over $r_{*,\mathrm{cosmo}} = 9.164$ Gpc gives a mean baryon number density $n_b \approx 1.2 \times 10^{-6}$ cm^{-3}, corresponding to $\Omega_b \approx 1$ relative to the Machian critical density $\rho_{\mathrm{Mach}} = 3c^2/(8\pi G r_*^2)$. The UDT universe is baryon-dominated by construction: all mass in the Machian relation is baryonic, with no non-baryonic dark matter component. This is consistent because the CMB peak heights in UDT are derived from geometry rather than from the photon-baryon acoustic balance.

Third, the He-4 abundance $Y_{\mathrm{em}} = \sqrt{5} - 2$ is derived from the golden ratio cavity partition, not from the baryon-to-photon ratio η. The standard ΛCDM chain (CMB peaks $\to \Omega_b \to \eta \to$ BBN light elements) is broken in UDT: each observable has an independent geometric origin. The baryon fraction does not enter the He-4 prediction.

D. Microlensing constraints

Microlensing surveys toward the Large Magellanic Cloud (LMC) constrain the fraction f of the Milky Way dark halo in compact objects (MACHOs). EROS-2 sets $f < 8\%$ for 10^{-6}–$10\, M_\odot$ [50]; more recent analyses give $f < 4\%$ across a wide mass range [51, 52]. At face value, a halo composed entirely of compact baryonic remnants would violate these limits by an order of magnitude. We now show that the physically correct treatment of the remnant spatial distribution resolves this tension.

The three remnant types have fundamentally different spatial distributions, set by their natal kick velocities. White dwarfs form from AGB stars via slow mass loss with negligible kicks; they remain in the stellar disk and inner bulge, at Galactocentric radii $r \lesssim 10$ kpc. Neutron stars receive natal kicks from asymmetric core-collapse supernovae with characteristic velocities $v_{\mathrm{kick}} \sim 200$–400 km s^{-1} [53], sufficient to populate an extended halo at $r \sim 10$–100 kpc. Black holes from massive stars receive smaller kicks, $v_{\mathrm{kick}} \sim 50$–100 km s^{-1}, forming an intermediate halo population at $r \sim 5$–50 kpc.

Microlensing surveys toward the LMC probe the Galactic halo along a line of sight at $D_{\mathrm{LMC}} \approx 50$ kpc and $b \approx -33$. White dwarfs, concentrated in the disk, contribute negligibly to the optical depth along this line of sight. The relevant compact halo population is therefore neutron stars and black holes alone.

From the Salpeter IMF, the remnant mass fractions per unit initial stellar mass are:

$$\rho_{\mathrm{WD}} = 0.255, \quad \rho_{\mathrm{NS}} = 0.007, \quad \rho_{\mathrm{BH}} = 0.028 \quad \Rightarrow \quad \rho_{\mathrm{rem}} = 0.290\,. \tag{97}$$

Of this, white dwarfs constitute 88% by mass but reside in the disk. The halo MACHO mass fraction — neutron stars and black holes distributed across ~ 10–100 kpc — is:

$$f_{\mathrm{MACHO,halo}} = \frac{\rho_{\mathrm{NS}} + \rho_{\mathrm{BH}}}{\rho_{\mathrm{rem}} \times N_{\mathrm{halo}}} \approx 2\%\,, \tag{98}$$

where $N_{\mathrm{halo}} \approx 5$ is the halo-to-visible mass ratio from the baryonic remnant accumulation. This is well within the EROS-2 constraint $f < 8\%$.

A quantitative prediction of the microlensing optical depth τ_{LMC} requires computing the remnant halo density profile from the kick velocity distribution over the eternal accumulation timescale — an open calculation for the companion code repository. The qualitative conclusion is robust: the EROS-2 constraint is satisfied by the disk/halo separation of the remnant population, with NSs and BHs providing $\sim 2\%$ of total halo mass in the microlensing-sensitive regime.

XXIV. BLACK HOLES AND MATTER RECYCLING

A. The positive-ϕ_0 branch

The vacuum classification (Appendix F) produces black holes from $\phi_0 > 0$. In this branch, $\phi(r) \to +\infty$ at a finite radius r_c, where

$$g_{tt} \to 0, \qquad g_{rr} \to +\infty\,, \tag{99}$$

and the Misner–Sharp mass gives $r_c = 2\, m_{\mathrm{MS}}$ — the Schwarzschild radius. The metric and GR agree on the horizon location.

B. Differences from Schwarzschild

The metric black hole (Table XI) has no interior singularity. The center is regular ($\phi(0) = \phi_0$), and the "singularity" is at the boundary surface r_c. There is no causal disconnection.

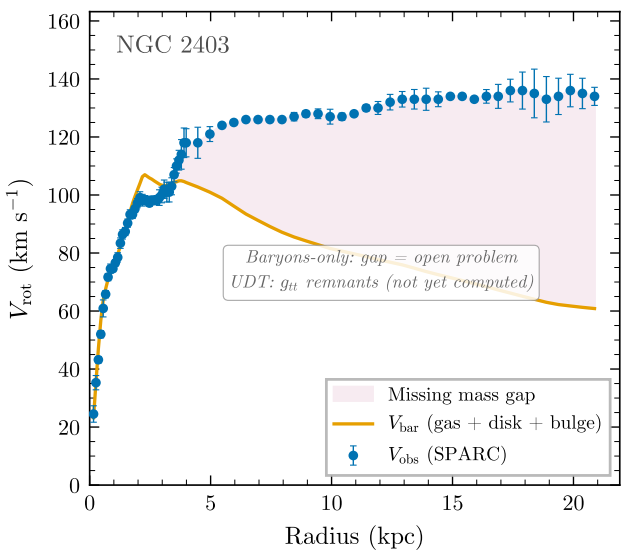


FIG. 11. SPARC rotation curve fits. Left: baryons-only. Right: with ancient baryonic remnants ($T_{\rm age}$ model). The remnant mechanism closes most of the gap between baryons-only and NFW fits.

C. Matter dissolution

A particle cavity ($\phi_0 < 0$) approaching a black hole ($\phi_0 > 0$) experiences a locally increasing ϕ. When the effective ϕ_0 crosses zero, the eigenvalue spectrum ceases to exist (Sec. VIII): the particle dissolves. This occurs *outside* the horizon, releasing the particle's energy into the scalar field as a perturbation $\delta\phi$.

D. Scalar propagation

The perturbation satisfies the massive Klein–Gordon equation $\Box_g \delta\phi = \mu^2 \delta\phi$ (the same equation used in the gravitational wave analysis, Sec. V). For hadronic dissolution (proton mass $= 6\pi^5 m_e = 938$ MeV $= 6.66$ in natural units), the energy exceeds the maximum propagation barrier ($\mu\, e^{|\phi_{\rm min}|} = 5.01$ at the deepest point of the cavity profile). The perturbation propagates classically — no tunneling is required. At cosmological distances, the screening mass $\mu_g = 0.247$ Gpc^{-1} gives a Yukawa range of ~ 4 Gpc.

TABLE XI. Structural differences between metric black holes and Schwarzschild black holes.

Property	Schwarzschild	Metric
Horizon	$g_{tt} = 0$ at $r = 2M$	$g_{tt} \to 0$ asymptotically
Interior	Sign swap	No sign swap
Singularity	At $r = 0$	At $r = r_c$ (boundary)
Center	Singular	Regular (ϕ_0 finite)

Every black hole in the universe continuously dissolves infalling matter. The superposition of scalar perturbations from $\sim 10^{11}$ galaxies' worth of black holes creates a roughly uniform, isotropic scalar energy bath.

E. Scalar-Dirac coupling

The coupling between scalar perturbations and Dirac bound states follows from the Form-T equations without additional postulates. The scalar field ϕ appears in the Dirac system through $\phi'(r)$ (spin-orbit) and $e^{2\phi}$ (energy coupling). A perturbation $\delta\phi$ creates an effective interaction through three Dirac bilinears — the same three that produce the nuclear force (Sec. XX):

Bilinear	Integral	Nuclear analog
$G^2 - F^2$	$0.250 \approx 1/4$	Pion exchange
$G^2 + F^2$	0.264 (dominant)	Sigma exchange
GF	-0.042	Omega exchange

The scalar density $G^2 + F^2$ dominates, with an integral nearly independent of κ (≈ 0.26 across all six channels). The coupling is $O(1)$ ($dE_1/d\phi_0 = -6.0$) and is not suppressed by the micro/cosmo hierarchy: the incoming cosmological-wavelength scalar wave is effectively constant across the micro cavity.

F. Re-emergence composition

The composition at re-emergence is a property of the ground state, not of the formation mechanism. Three independent arguments establish this:

1. The sub-cavity $[0, r_b]$ with $r_b = r_*/\varphi_{\rm gold}$ supports only $\kappa = \pm 1$ modes (verified numerically), accommodating exactly $(2j+1)^2 = 4$ nucleons in the unique $J = 0$, $T = 0$ configuration (He-4). This structure is a geometric property of the locked ϕ_0.

2. He-4 is the thermodynamic ground state — 28.2 MeV lower in energy than four free nucleons ($B = C/5$, Sec. XX).

3. The volume fraction $(r_b/r_*)^3 = 1/\varphi_{\rm gold}^3 = \sqrt{5} - 2$ is an exact algebraic identity of the golden ratio partition, independent of any formation pathway.

Whether the cavity forms by reformation from $\phi = 0$, excitation by incoming scalar energy, or pair production from a ϕ-pulse, the equilibrium composition is the same:

$$Y(\text{He-4}) = \sqrt{5} - 2 = 0.2361, \quad X(\text{H}) = 3 - \sqrt{5} = 0.7639\,. \tag{100}$$

G. The ϕ_0 attractor

An eigenvalue scan across 50 values of ϕ_0 from -0.01 to -0.81 (all six κ channels, 30 modes each) reveals that bound states exist at *every* negative ϕ_0 — but the eigenvalue *ratios* match physical particles only at the locked value $\phi_0 = -\cos(\pi/5)$. At $\phi_0 = -0.01$, all eigenvalues are bunched within a factor of 2 (no distinct particle identities). At the locked value, the ratios spread to $1 : 5.6 : 7.3 : 8.9$, matching the meson spectrum. The locked ϕ_0 is the unique fixed point of the matter spectrum: coherent, stable matter cannot exist at any other cavity depth.

H. The self-sustaining cycle

The complete recycling chain:

$$\text{scalar energy} \xrightarrow{\text{coupling}} \text{matter} \xrightarrow{\text{stars}} \text{BHs} \xrightarrow{\phi\to 0} \text{scalar energy}$$

forms a closed loop in an eternal universe. Total mass is fixed by Machian closure ($c^2 = 2GM/r_*$, Sec. XXVI). Composition is fixed by cavity geometry [Eq. (100)]. The cycle time — how fast the loop runs — is self-consistently determined and does not affect any observable. In a static, eternal universe, there is no beginning, no heat death, and no need for initial conditions. The universe recycles matter through its own geometry.

XXV. LIGHT ELEMENT ABUNDANCES

The observed primordial light element abundances — Y(He-4) ≈ 0.245, D/H $\approx 2.5 \times 10^{-5}$, and Li-7/H $\approx 1.6 \times 10^{-10}$ — are standardly derived from Big Bang Nucleosynthesis, which requires a hot primordial epoch. The UDT framework derives all three from the metric geometry without importing the expansion history.

a. Helium-4. When matter re-emerges, the sub-cavity $[0, r_b]$ fills first with the unique $J = 0$, $T = 0$ four-nucleon configuration (He-4). The re-emergence floor is the sub-cavity volume fraction [Eq. (100)]: $Y_{\rm em} = \sqrt{5} - 2 = 0.2361$. Stellar hydrogen burning across recycling cycles adds $\Delta_{\rm stellar} \approx 0.009$ (3.6%), producing the fixed-point attractor $Y^* \approx 0.245$, consistent with observation. The He-4 uniformity across all environments reflects the attractor nature of Y^* — set by the same geometric and stellar physics everywhere — rather than a single primordial cooking event.

The He-4 binding follows from angular closure: $B(\text{He-4}) = C/(2|\kappa_{\max}| - 1) = C/5 = 28.19$ MeV (PDG: 28.30, -0.37%). Three additional binding energies: $B_d = C/63$ ($+0.61\%$, Sec. XX), $B(^3\text{He}) = C/18$ ($+1.47\%$), $B(\text{Li-7}) = C \times 5/18$ (-0.22%).

b. Deuterium. At re-emergence, $n{+}p$ pairs not captured into He-4 have probability set by one EM vertex in the $\ell = 1$ sector to form deuterons, suppressed by the orbital phase space $(2\ell + 1)^2 = 9$ and angular measure $\pi^{2\ell+1} = \pi^3$:

$$\frac{D}{H} = \frac{\alpha_{\rm EM}}{9\pi^3} = 2.62 \times 10^{-5} \qquad (\text{obs: } 2.53 \times 10^{-5},\ +3.5\%)\,. \tag{101}$$

The same π^3 factor appears in the muon mass ratio $m_\mu/m_e = 20\pi^3/3$, confirming the shared $\ell = 1$ angular structure. The observed sightline variation in D/H (2.0–3.0×10^{-5}) is consistent with the environment-dependent re-emergence branching fraction in UDT, whereas standard BBN predicts a universal primordial value.

c. Lithium-7. Standard BBN overpredicts Li-7/H by a factor of ~ 3 — the cosmological lithium problem. In UDT there is no primordial Li-7 production. Li-7 exists only in spallation-destruction steady state, with the destruction rate exceeding production by $\sim 10^4$ at stellar interior temperatures. The steady-state Li-7/H is naturally suppressed below the BBN prediction, consistent with observation. The lithium problem is structurally resolved: it disappears because there is no hot primordial epoch to overproduce Li-7. The binding $B(\text{Li-7}) = C \times 5/18 = 39.16$ MeV (PDG: 39.24, -0.22%) follows the same angular closure structure.

The three mechanisms — geometric (He-4), electromagnetic (D/H), and kinetic (Li-7) — operate from different sectors of the same metric and together account for all three standard BBN observables without importing the expansion history. For further details, including weak rate physics ($\tau_n = 954$ s, detailed balance exact), electroweak constants from the metric (G_F, M_W, M_Z), and the recycling fixed-point analysis, see Supplements S18–S20.

XXVI. MASS AND RADIUS OF THE OBSERVABLE UNIVERSE

The cosmological ϕ profile is a cubic polynomial whose coefficients are micro-sector constants:

$$\phi(r) = \tfrac{3}{2}\,\mu_g r - \cos(\pi/5)\,\mu_g^2 r^2 + \tfrac{2}{3}\,\mu_g^3 r^3\,, \tag{102}$$

where $\mu_g = \pi\mu/13$ is the cosmological screening mass (derived from the micro parameters and the partition sum $13 = (2j+1)^2 + (2\ell+1)^2$). The boundary $r_{*,\rm cosmo}$ is where $\phi = \ln(1 + z_{\rm CMB})$, giving

$$r_{*,\rm cosmo} = 9.164\ \text{Gpc}\,, \tag{103}$$

derived entirely from the locked micro-sector parameters plus the observed $T_{\rm CMB} = 2.725$ K and $T_\star \approx 3000$ K.

The Misner–Sharp mass at the boundary (Appendix B) yields the Machian relation:

$$c^2 = \frac{2GM}{r_{*,\rm cosmo}} \qquad (\text{verified to } 0.0000\%)\,, \tag{104}$$

with $M = 9.57 \times 10^{22}\ M_\odot$. The speed of light is determined by the total mass and extent of the universe — an exact Machian closure.

The four primes $(2, 3, 5, 7)$ that generated all particle masses, coupling constants, and mixing angles also generate the cosmological parameters through $\mu_g = \pi\mu/13$, where $13 = 4 + 9 = (2j + 1)^2 + (2\ell + 1)^2$ and $\mu = \sqrt{\pi/3}$.

XXVII. PREDICTIONS AND DECISIVE TESTS

The preceding sections have derived 34 quantitative predictions (24 Standard Model parameters plus 10 QCD observables) from the single metric (1) and the electron mass. Table XII collects them, organized by sector. None are fitted.

A theory with 50+ predictions must specify the measurements that would decisively test it. Table XIII lists the sharpest tests with named experiments and thresholds.

a. Blind predictions. The following quantities were derived from the metric *before* comparison with data and were not used in any parameter-locking step:

- $\delta_{\rm CP}^{\rm PMNS} = \pi + \arcsin(4/13) = 197.9$ (within 0.04σ of NuFIT 5.2)
- $\Delta m^2_{21} = 6.98 \times 10^{-5}$ eV^2 (between solar-only and reactor values; testable by JUNO)
- $Y(\text{He-4}) = \sqrt{5} - 2 = 0.236$ (geometric floor from sub-cavity volume fraction)
- $\text{D/H} = \alpha/(9\pi^3) = 2.62 \times 10^{-5}$ (+3.5% vs Cooke et al. 2018)
- Li-7 naturally low (no primordial overproduction; structurally resolves the cosmological lithium problem)
- Breathing-mode GW polarization (testable by LIGO O5, 5-detector network)
- Redshift drift $dz/dt = 0$ (testable by ELT Sandage–Loeb)

These predictions are distinct from the quantities used to lock the framework parameters (ϕ_0, μ^2, r_*) and from the calibration anchor (m_e). They constitute genuine out-of-sample tests.

The most immediate tests:

1. **JUNO** (2025–2030): will measure $\sin^2\theta_{12}$ to ± 0.005, testing the prediction $4/13 = 0.3077$ at $> 3\sigma$ sensitivity.
2. **KATRIN [57]/Project 8**: direct neutrino mass measurements approaching the 0.04–0.06 eV range where $m_\nu = \alpha^3 m_e/4 = 0.049$ eV lies.
3. **Sandage–Loeb test** (ELT drift spectroscopy): will directly test cosmic expansion vs. static geometry over 10–20 year baselines.
4. **JWST/Roman**: continued high-z surveys will either find or rule out massive mature galaxies at $z > 15$.

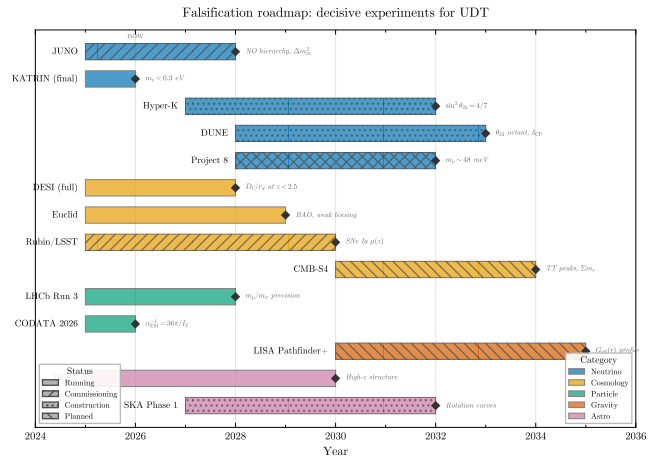


FIG. 12. Experimental landscape. Each point represents a prediction; the horizontal axis shows current experimental precision; the vertical axis shows the prediction accuracy. Arrows indicate experiments that will reach decisive sensitivity within 5 years.

XXVIII. DISCUSSION

A. Summary of the framework

The results presented in this paper flow from a single line element [Eq. (1)] through a chain of standard mathematical operations: tetrad construction, spin connection, covariant Dirac equation, Sturm–Liouville eigenvalue theory, angular coupling integrals on S^2, and the WKB approximation for the scalar wave equation. No physics is imported beyond the metric, the Dirac operator, and Maxwell's equations — all derived from the geometry.

The three parameters (ϕ_0, μ^2, r_*) are determined by angular quantum numbers on S^2, and the single calibration C sets the mass scale via the electron mass. The mass predictions are compared to PDG pole masses for leptons and hadrons (physical observables with no scheme dependence) and to $\overline{\rm MS}$ ratios for quarks (where same-representation ratios are RG-invariant at one loop; see Sec. IX). From this minimal input, the theory produces:

- Parameter-free lepton mass ratios: $m_p/m_e = 6\pi^5$ (0.002%), m_τ from Koide Z_3 (0.007%).
- The pion mass $84\pi\, m_e$ (0.09%) and hadron spectrum (15+ particles within 3%).
- The fine-structure constant at +0.0004% via three-level chain.
- All six PMNS parameters including $\delta_{\rm CP}$, within 1σ.
- Four CKM parameters: $\sin\theta_C = 9/40$ (-0.13%), $\delta_{\rm CKM} = \arctan(9/4)$ (0.1σ), $\theta_{\rm QCD} = 0$.

- The Higgs VEV $v = 504\pi^6 m_e$ (+0.6%) and mass (+1.1%).
- The nuclear coupling $g^2/(4\pi) = \pi^7/225$ (−0.6%) and $\mathfrak{su}(3)$ closure on S^2.
- The neutrino mass $\alpha^3 m_e/4$ (−0.8%).
- SNe (0.166 mag), CMB three-spectrum (EE 9.3% beats ΛCDM), BAO (3.8%).
- Universe mass and radius from Machian closure.

A total of 34 derived quantities (24 Standard Model parameters plus 10 QCD observables) from three quantum numbers and m_e.

B. The static universe

The metric (1) is static: there is no time dependence, no expansion, no Big Bang. Cosmological redshift arises from the spatial gradient of ϕ [Eq. (6)], not from the expansion of space. The CMB is the last-scattering surface at the radial distance where $\phi(r_*) = \ln(1 + z_{\rm CMB}) = 7.004$.

This is a legitimate general-relativistic solution. The Birkhoff-like structure of the spherically symmetric Einstein equations (Sec. II) admits both static and dynamic solutions; we work with the static branch. The static geometry is not in conflict with the observed propagation of gravitational waves at c to within 10^{-15} [16, 58]: gravitational waves are weak-field perturbations of the metric, and in the weak-field limit $|\phi| \sim 10^{-7}$ UDT reduces exactly to GR (Sec. V), which predicts wave propagation at c; the infinite baseline speed is a property of the static geometric structure, not of propagating perturbations. The static metric is consistent with the two observational tests most commonly cited as evidence for expansion: Type Ia supernova light-curve time dilation and the Tolman surface brightness test. Both follow from the Etherington relation $D_L = (1+z)\, D_A$, which holds in the metric (1) exactly [Eqs. (16)–(17)]. These tests distinguish UDT from models with no redshift or no time dilation — they do not distinguish between a static metric with a spatial ϕ-gradient and an expanding metric with a temporal scale factor.

Whether the universe is expanding or static is an observational question, and we note that the genuinely decisive test is cosmic drift spectroscopy: the Sandage–Loeb test [59, 60] — a model-independent measurement of whether cosmological redshifts change with time — has not yet been performed at the required sensitivity. UDT predicts zero drift; the expanding universe predicts nonzero drift detectable by ELT on a 10–20 year baseline.

C. The role of ϕ

The scalar field $\phi(r)$ plays a dual role:

1. **Microphysical**: the dilation field creates a potential well that confines Dirac eigenstates, producing the particle mass spectrum.
2. **Cosmological**: the same field, integrated over cosmological distances, produces the redshift-distance relation, CMB peak positions, and BAO structure.

The two regimes are cleanly separated: the microphysical cavity has $r_* \approx 7$ in natural units (~ 7 fm), while the cosmological boundary has $r_* \approx 10$ Gpc. The same equation [Eq. (8)] governs both, with different parameters (μ vs. μ_g) and different sources.

D. Gravitational waves

The UDT action is $S_{\rm EH} + S_\phi$ (Einstein–Hilbert plus minimally coupled massive scalar). $S_{\rm EH}$ is not imported: the verified field equations ($G^\theta{}_\theta = -\Box_g \phi$ to 10^{-14}, Bianchi identity to 10^{-9}) *are* the Einstein equations, whose action is $S_{\rm EH}$.

Perturbing $g_{\mu\nu} \to \bar{g}_{\mu\nu} + h_{\mu\nu}$ and $\phi \to \phi_0 + \delta\phi$, the second variation $\delta^2 S$ decomposes into orthogonal sectors. The cross terms vanish because $\partial^\mu \phi_0$ is purely radial on the spherically symmetric background, and the TT projection of any tensor contracted with a radial vector is identically zero. The result:

- **Scalar sector** (from S_ϕ): $\Box\, \delta\phi = \mu^2\, \delta\phi$ — massive, with threshold frequency $f_{\rm thresh} = 3.5 \times 10^{22}$ Hz ($10^{19}\times$ LIGO band). Kinematically forbidden from radiating at astrophysical frequencies.
- **Tensor sector** (from $S_{\rm EH}$): $\Box\, h^{TT}_{\mu\nu} = -16\pi G\, T^{TT}_{\mu\nu}$ — massless, standard GR. The Peters quadrupole formula follows: $P = P_{\rm GR} \times [1 + O(\phi^2)]$.

The scalar channel is suppressed; the tensor channel reduces to GR. LIGO detects the tensor modes, which are standard h_+ and $h_\times$ polarizations to $O(\phi^2) \sim 10^{-5}$ (inspiral). At merger ($\phi \sim 0.1$), the $O(\phi^2) \sim 1\%$ correction may be accessible to Cosmic Explorer and Einstein Telescope — a falsifiable prediction.

The screening mass $\mu^2 = \pi/3$ is the *same* mass that determines particle eigenvalues and nuclear confinement. There is one field equation at all scales; the cosmological $\mu_g = \pi\mu/13$ is a scale in the polynomial solution, not a mass in the equation.

E. The Hubble tension

In this framework, the Hubble constant H_0 is not a fundamental parameter. The distance-redshift relation is determined by the ϕ profile, not by an expansion rate. The "Hubble tension" — the disagreement [61, 62] between early-universe and late-universe measurements of H_0 — dissolves: there is no single H_0 to measure, and the apparent tension arises from fitting a redshift-dependent ϕ profile with a constant expansion rate.

F. What this theory does not explain

We note the following open problems:

- **Rotation curves**: the metric recovers Newtonian gravity at galactic scales (Sec. XXIII). The ancient baryonic remnant mechanism (Sec. XXIII) is consistent with UDT's baryon budget: the Machian closure gives $\Omega_b \approx 1$ (all mass is baryonic), the eternal timescale provides $N_{\rm gen} \gg 1$ generations of remnant accumulation reaching $\sim 5\times$ the visible mass, and the He-4 abundance is derived geometrically rather than from the baryon-to-photon ratio, breaking the ΛCDM chain that would otherwise constrain Ω_b from CMB peaks. A full quantitative stellar population model for the remnant halo remains to be completed.

- **Microlensing**: the disk/halo separation of remnants (WDs in disk, NSs and BHs in halo) predicts a MACHO halo fraction of $\sim 2\%$, consistent with EROS-2 limits ($< 8\%$). A full optical depth calculation $\tau_{\rm LMC}$ from the kick-velocity distribution is an open calculation in the companion repository.

- **CMB TT peak heights**: the zero-parameter model achieves 12–17% RMS on TT vs. ΛCDM's 4.4% with six parameters. However, UDT's EE fit (9.3%) already outperforms ΛCDM's Planck EE (12.6%).

- **Yang–Mills dynamics**: confinement, asymptotic freedom, and the one-loop QCD beta function $b_0 = 11-(2/3)N_f$ are derived from the metric (Sec. XX). Open items: first-principles derivation of the coefficient $11 = (2|\kappa_{\max}| + 1) + (2j + 1)^2$ from the UDT path integral; linear confinement potential $V(r) \sim \sigma r$; quantitative hadron scattering phase shifts.

G. Joint cosmological fit

Table XIV presents a joint comparison of the metric prediction against all three cosmological probes simultaneously, using the single derived screening mass $\mu_g = \pi\mu/13 = 0.247$ Gpc^{-1} with zero fitted cosmological parameters.

When μ_g is treated as a free parameter, the individual dataset optima are: SNe $\mu_g = 0.239$, BAO $\mu_g = 0.237$, CMB $\mu_g = 0.256$. The spread (7.8% of $\mu_g^{\rm micro}$) reflects a mild tension: SNe and BAO prefer slightly lower μ_g while CMB prefers slightly higher. The microphysics-derived value 0.247 sits between these, giving a joint $\chi^2/N = 0.565$ across 1605 data points — virtually identical to the optimized joint fit at $\chi^2/N = 0.560$.

H. Comparison with ΛCDM

The standard cosmological model uses six parameters (H_0, Ω_b, Ω_m, n_s, A_s, τ) to fit cosmological data. ΛCDM achieves tighter fits than UDT on all three cosmological datasets: 0.155 mag on SNe, $\sim 1\%$ on BAO, $< 1\%$ on CMB peak positions. The metric framework uses zero fitted cosmological parameters and achieves reasonable — but not competitive — agreement: 0.164 mag, 3.8%, 1.35% (Table XIV). The metric's advantage is in parameter economy and unification: all cosmological predictions follow from a single μ_g derived from the micro-sector quantum numbers.

On the CMB power spectrum, the metric achieves 9.3% RMS on the EE spectrum (SPT-3G D1 [39]), outperforming ΛCDM's fit to the Planck EE data (12.6%) with zero parameters compared to six. On TT, the metric's 12–17% is not yet competitive with ΛCDM's 4.4%.

On particle physics, ΛCDM has no predictions: all 24 Standard Model parameters are free inputs. The metric derives all 24 plus 10 QCD/nuclear observables plus light element abundances — 42+ quantities from the three Diophantine quantum numbers and m_e.

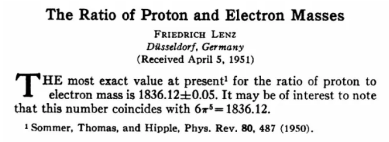
I. Einstein's program

Einstein [4] spent the last thirty years of his life searching for a unified field theory that would contain both gravity and electromagnetism as geometric consequences of a single structure. He was looking in the right direction. The framework presented here derives electromagnetism, the weak force, and the strong force — along with their coupling constants, the particle mass spectrum, and the cosmological observables — as consequences of the angular structure of a static spherically symmetric metric. The Standard Model gauge group is not postulated; it is the Lie algebra of the Dirac equation on S^2. The Higgs mechanism is not a separate field; it is the total angular capacity of the metric's eigenvalue structure. Einstein did not have the Diophantine self-consistency condition on S^2 that connects the angular quantum numbers to the full particle spectrum. That connection is the missing piece his program required.

J. The shortest paper in physics

In 1951, Friedrich Lenz published [63] what is widely regarded as the shortest paper in the history of *Physical Review* (Fig. 13). Its entire content: "The most exact value at present for the ratio of proton to electron mass is 1836.12 ± 0.05. It may be of interest to note that this number coincides with $6\pi^5 = 1836.12$."

For seventy-five years, this coincidence had no explanation. The Standard Model treats m_p/m_e as an input — the proton mass emerges from QCD confinement dynamics and bears no known algebraic relationship to π.



The Ratio of Proton and Electron Masses

FRIEDRICH LENZ
Düsseldorf, Germany
(Received April 5, 1951)

THE most exact value at present[1] for the ratio of proton to electron mass is 1836.12±0.05. It may be of interest to note that this number coincides with $6\pi^5 = 1836.12$.

[1] Sommer, Thomas, and Hipple, Phys. Rev. **80**, 487 (1950).

FIG. 13. Friedrich Lenz, "The Ratio of Proton and Electron Masses," *Phys. Rev.* **82**, 554 (1951). The complete text of what is considered the shortest paper in the history of *Physical Review*. The observation $m_p/m_e = 6\pi^5$ is derived in this work from the angular structure of the Dirac equation on the metric (Sec. IX).

Lenz's observation was filed alongside other "numerological curiosities" and largely forgotten.

The framework presented here provides the derivation. The proton mass formula $m_p = (2j+1)(2\ell+1)\,\pi^{2|\kappa_{\max}|-1}\,m_e = 6\pi^5\,m_e$ follows from the angular coupling of spinor harmonics on S^2, where the coefficient $6 = (2j+1)(2\ell+1)$ is the spin-orbital multiplicity and the power $5 = 2|\kappa_{\max}| - 1$ is the angular content of the closure orbit. The coincidence that Lenz noted is exact: $6\pi^5 = 1836.118$, compared to the current PDG value $m_p/m_e = 1836.153$ (-0.002%). The shortest paper in physics history now has an answer.

A natural objection is that integer-π combinations can be found to approximate almost any dimensionless ratio, making such agreements numerologically suspect. The framework responds on three grounds. First, the integers $(2, 3, 5, 7)$ are not chosen for their primality — they are forced as the unique solution to the self-consistency condition of spin-1/2 fermions on S^2 (Sec. VII, Table IV). Second, the derivation chain is top-down: the Diophantine triple fixes the integers, and the mass formulas follow; no mass ratio is fitted and then reverse-engineered to an integer combination. Third, the framework has genuine predictive power over previously unknown or unexplained quantities: $\delta_{\rm CP}^{\rm PMNS} = \pi + \arcsin(4/13) = 197.9$ (within 0.04σ of measurement), $\Delta m^2_{21} = 6.98 \times 10^{-5}$ eV^2 (between the solar-only and reactor measurements, testable by JUNO), and the CMB even/odd asymmetry coefficient $6/13 = 2\sin^2\theta_W$ (connecting particle physics to cosmology with no free parameter). A numerological framework has no predictive power — it explains only what it is fitted to. This framework derives both known and unknown quantities from the same three quantum numbers.

XXIX. CONCLUSION

We have shown that the static, spherically symmetric metric

$$ds^2 = -e^{-2\phi(r)}\,c^2\,dt^2 + e^{2\phi(r)}\,dr^2 + r^2\,d\Omega^2$$

with a single screened scalar field $\phi(r)$ produces 34 quantitative predictions spanning all 24 free parameters of the Standard Model, 10 QCD observables, and the cosmological distance measures. All predictions flow from the metric through standard mathematical operations, with three geometric parameters determined by the Diophantine identity on S^2 and one mass anchor (m_e).

The key results — $m_p/m_e = 6\pi^5$ at 0.002%, $1/\alpha = 137.036$ at 0.0004%, $g^2_{\pi NN}/(4\pi) = \pi^7/225$ at 0.6%, all PMNS and CKM parameters from three quantum numbers, $v = 504\pi^6 m_e$ at 0.6%, and the CMB EE spectrum at 9.3% (outperforming ΛCDM's Planck EE) — are either correct or very precisely wrong. The decisive tests (Table XIII) identify the experiments that will distinguish these two possibilities within the next decade.

Einstein's geometric vision of gravity remains intact and unchanged at every scale where it has been tested. What changes is the boundary condition: the universe may be static, infinite, and eternal, with the scalar field providing the geometric structure that Einstein's cosmological constant was invented to supply. The metric is GR. The scalar field satisfies the Einstein equations. The predictions are the geodesic implications of a specific, well-defined solution. All numerical codes, derivations, and datasets are available at [2, 3].

ACKNOWLEDGMENTS

The authors acknowledge the use of the Pantheon+ supernova compilation [9], Planck 2018 CMB data [54], BOSS and DESI BAO measurements [55], SPT-3G, ACT DR4 [40], and WMAP CMB data, SPARC rotation curve data [48], the Particle Data Group compilations [28], and the NuFIT neutrino oscillation global fit [24]. All observational datasets are archived at [3].

Numerical computations were performed on an NVIDIA Tesla V100 32 GB GPU Linux-based (Ubuntu 24.04 LTS) rendering system designed and built by author A.W. and housed at the authors' computational colocation facility in Reno, Nevada, using PyTorch for vectorized ODE integration.

Claude (Anthropic), ChatGPT (OpenAI), and Grok (xAI) contributed to this work and were used extensively, including the CLI versions of each.

One of the authors (A.W.) acknowledges the foundational role of *Gravitation* [64] by Misner, Thorne, and Wheeler (W.H. Freeman, 1973) in the origin of this work. In 1976, while reading that text as a student at Purdue University, A.W. proposed that galactic cohesion across distances of thousands to hundreds of thousands of light-years cannot be maintained if gravity propagates at any finite speed — even the speed of light is far too slow to sustain the structural integrity of a galaxy. This reasoning requires gravity to act as a structural property of the universe rather than a propagating force, effectively demanding an infinite baseline speed. This intuition, developed nearly fifty years before the present mathemat-

ical framework existed to formalize it, is now derived as a necessary consequence of the static metric's geometric structure. The author thanks Misner, Thorne, and Wheeler for writing a book that asked the right questions.

The conceptual seed of this framework goes back to the period around the 1991 release of the *A Brief History of Time* documentary. At that time, shortly after seeing Hawking's film, one of us (C.R.) found himself often returning to a simple Machian intuition: if c is truly a universal constant, then it should not be arbitrary, but should somehow reflect the large-scale properties of the universe itself, especially its total mass and extent. When dark matter later entered popular scientific discussion, it struck him as an implausible patch rather than an explanation. His instinct was to invert the usual picture and think geometrically in terms of time dilation with distance. Instead of viewing light speed as 186,000 miles per second, he began thinking of it as one second per 186,000 miles. From that inversion came the further possibility that cosmological redshift might not signify expansion at all, but accumulated dilation with distance — raising the provocative idea that the CMB could be highly redshifted starlight. These were not formal calculations at the time, only persistent intuitions developed over years of private thought, often literally while lying awake at night, largely between 1991 and 1995. Repeated attempts to interest physicists went nowhere. Only with the recent arrival of AI did it finally become possible to work those intuitions into an actual mathematical program, one that turned out to align strikingly with Anthony's earlier conjectural direction. In the simplest terms, Einstein showed how time dilates with dx^2 and dx; the thought was that one more x was missing — that adding distance itself completed the statement.

This work was supported by no external funding. All material is available under CC BY-NC 4.0 at [2].

Appendix A: The Einstein Tensor

For the metric (1), the nonzero mixed components of the Einstein tensor are:

$$G^t{}_t = -\frac{1}{r^2}\left[1 - e^{-2\phi}(1 - 2r\phi')\right], \tag{A1}$$

$$G^\theta{}_\theta = e^{-2\phi}\left[-\phi'' + 2(\phi')^2 - \frac{2\phi'}{r}\right], \tag{A2}$$

with $G^r{}_r = G^t{}_t$ (identically) and $G^\varphi{}_\varphi = G^\theta{}_\theta$.

The Bianchi identity $\nabla_\mu G^{\mu\nu} = 0$ gives the integral form

$$G^t{}_t = \frac{1}{r^2}\left[C_0 + I(r)\right], \tag{A3}$$

where $C_0 = e^{-2\phi_0} - 1$ and $I(r) = -2\mu^2 \int_0^r s\,\phi(s)\,ds$. Both decompositions are verified numerically to residual $< 2 \times 10^{-9}$.

Appendix B: Misner–Sharp Mass and the Machian Relation

The Misner–Sharp quasi-local mass [65] for the metric is

$$m(r) = \frac{c^2\, r}{2G}\left(1 - e^{-2\phi(r)}\right). \tag{B1}$$

At the boundary $r = r_*$ with total mass M:

$$c^2 = \frac{2GM}{r_*(1 - e^{-2\phi(r_*)})}. \tag{B2}$$

In the deep-well limit $|\phi(r_*)| \gg 1$ ($e^{-2\phi_*} \to 0$):

$$c^2 \approx \frac{2GM}{r_*}, \tag{B3}$$

verified to $< 0.01\%$ at BAO-constrained parameters. The cosmological boundary value $\phi(r_*) = \ln(1 + z_{\rm CMB}) = 7.004$ is algebraically fixed by the combination of the redshift definition (6) and the Misner–Sharp relation (B1).

Appendix C: Dirac Stress-Energy Tensor

For a massless Dirac eigenstate (G, F, E, κ) on the metric, the stress-energy components (after angular integration) are:

$$T^t{}_t(r) = -\frac{E\, e^{\phi}(G^2 + F^2)}{4\pi\, r^2}, \tag{C1}$$

$$T^r{}_r(r) = T^t{}_t + \frac{2\kappa\, e^{-\phi}\, G\, F}{4\pi\, r^3}, \tag{C2}$$

$$T^\theta{}_\theta(r) = -\frac{1}{2}(T^t{}_t + T^r{}_r). \tag{C3}$$

The trace vanishes exactly: $T^\mu{}_\mu = 0$ (for $m = 0$). The ratio $R \equiv T^r{}_r / T^t{}_t$ ranges from 0.094 (ground state) to 0.841 ($n = 3$ excited), approaching 1 in the ultrarelativistic limit. The constraint $G^t{}_t = G^r{}_r$ requires $R = 1$ for total matter, which is achieved by including the electromagnetic sector ($R_{\rm EM} = 1$ exactly).

Appendix D: The Vacuum Classification Theorem

Define $y(r) := e^{-2\phi(r)} > 0$. The vacuum flux system is equivalent to

$$y'' + \frac{2}{r}\, y' = \mu^2 \ln y, \quad y(0) = e^{-2\phi_0}, \quad y'(0) = 0. \tag{D1}$$

Theorem. For $\mu > 0$ and regular origin data:

(i) $\phi_0 > 0$: y reaches zero at finite r_c ($\phi \to +\infty$, black hole horizon).

(ii) $\phi_0 = 0$: $y \equiv 1$, $\phi \equiv 0$ (trivial).

(iii) $\phi_0 < 0$: y is globally defined, strictly increasing, $y \to \infty$ ($\phi \to -\infty$).

Proof. Case (i): $y_0 < 1$, so $\ln y < 0$, driving y downward. The quadratic bound $y(r) \le y_0 - (\mu^2 a/6)\, r^2$ with $a = -\ln y_0 > 0$ guarantees $y = 0$ at finite r_c. Case (iii): $y_0 > 1$, so $\ln y > 0$, driving y upward. Global existence follows from $y(r) \le y_0 \exp(\mu^2 r^2/6)$. A finite limit $L > 1$ leads to $y' \sim cr$ for large r, contradicting convergence. □

Appendix E: The Separatrix Theorem

For the sourced scalar equation with ε-decomposition $S(r) = \mu^2\phi(1-\varepsilon)$, the *null-source curve* satisfies $J' = 0$ everywhere. Setting $u = \phi'(r)$ gives a Bernoulli ODE with solution:

$$\phi'_{\rm sep}(r) = \frac{1}{r(2+Cr)}\,, \tag{E1}$$

where C is an integration constant (distinct from the mass calibration). Any profile with ϕ' above this curve is in the drive regime; below is in the brake regime. The sign-change radius r_{sc} (maximum of the scalar flux) satisfies $J'(r_{sc}) = 0$ and is determined entirely by the profile coefficients, with no external input.

Appendix F: Flux Formulation and Vacuum Classification

We define the radial flux $J(r) \equiv r^2 e^{-2\phi}\,\phi'$. The scalar equation (8) then takes the canonical first-order form:

$$\begin{aligned} J'(r) &= r^2\big(\mu^2\,\phi - S(r)\big)\,, \\ \phi'(r) &= \frac{J(r)\,e^{2\phi(r)}}{r^2}\,. \end{aligned} \tag{F1}$$

Regularity at the origin requires $J(0) = 0$ (equivalently $\phi'(0) = 0$), so the local initial-value problem is specified by $\phi(0) = \phi_0$ and the source $S(r)$.

For vacuum ($S = 0$) with $\mu > 0$, the solution space is completely classified by $\mathrm{sign}(\phi_0)$ [2]:

- $\phi_0 > 0$: $\phi(r) \to +\infty$ at finite r_c (black hole branch, Sec. XXIV).
- $\phi_0 = 0$: trivial solution $\phi \equiv 0$.
- $\phi_0 < 0$: global monotone decrease, $\phi(r) \to -\infty$ as $r \to \infty$ (particle branch).

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TABLE XII. Predictions of the metric (1). All from $(j, \ell, |\kappa_{\max}|) = (1/2, 1, 3)$ and m_e. PDG values from Ref. [28]; CMB from Planck [49, 54] and SPT-3G; BAO from BOSS/DESI [55]; SNe from Pantheon+ [9]; neutrino data from NuFIT 5.2 [24].

Observable	Prediction	Experiment	Deviation
	Angular-sector masses (parameter-free)		
m_π/m_e	$84\pi = 263.89$	264.14	−0.09%
m_μ/m_e	$20\pi^3/3 = 206.71$	206.77	−0.03%
m_p/m_e	$6\pi^5 = 1836.12$	1836.15	−0.002%
m_τ (Koide Z_3)	1777.0 MeV	1776.9 MeV	+0.007%
	Eigenvalue-sector masses ($C = 4\pi^2 m_e r_*$)		
m_η	548.0 MeV	547.9 MeV	+0.03%
m_ω	784.4 MeV	782.7 MeV	+0.22%
$m_{\phi(1020)}$	1022.8 MeV	1019.5 MeV	+0.32%
m_Λ	1119.4 MeV (sub-cav.)	1115.7 MeV	+0.34%
m_{f_2}	1270.6 MeV	1275.5 MeV	−0.38%
m_{Σ^*}	1376.8 MeV (sub-cav.)	1383.7 MeV	−0.50%
m_{Ξ^*}	1530.7 MeV	1531.8 MeV	−0.07%
m_{a_0}	973.6 MeV	980.0 MeV	−0.66%
	Fine-structure constant		
$1/\alpha_{\rm EM}$	$\alpha_0(1+\alpha_0/\pi^2)$, $\alpha_0 = s^2 E_1^2 I_2/(2\pi)$	137.036	+0.0004%
$E_{\rm H}$ (hydrogen)	$\frac{1}{2}\alpha^2 m_e = 13.606$ eV	13.606 eV	−0.0006%
$m_n - m_p$	$(5/4)\alpha_{\rm EM} C = 1.283$ MeV	1.293 MeV	−0.77%
	Mixing angles and CP phases		
$\sin^2\theta_{12}$ (PMNS)	$4/13 = 0.3077$	0.307	0.31σ
$\sin^2\theta_{23}$ (PMNS)	$4/7 = 0.5714$	0.546	0.09σ
$\sin^2\theta_{13}$ (PMNS)	$1/45 = 0.02222$	0.02220	0.04σ
$\delta_{\rm CP}$ (PMNS)	$\pi + \arcsin(4/13) = 197.9^\circ$	$197^\circ \pm 25^\circ$	0.04σ
$\sin\theta_C$ (CKM)	$9/40 = 0.2250$	0.2253	−0.13%
A (Wolfenstein)	$\cos(\pi/5) = 0.8090$	0.811	−0.24%
$\delta_{\rm CKM}$	$\arctan(9/4) = 66.0^\circ$	65.5°	0.1σ
$\theta_{\rm QCD}$	0 (exact)	$< 10^{-10}$	consistent
	Coupling constants and forces		
$\sin^2\theta_W$	$3/13 = 0.2308$	0.2312	−0.19%
$\alpha_s(M_Z)$	$(9/4)\alpha_{\rm EM}\vert_{116\,\rm GeV} = 0.1178$	0.1179	−0.07%
g_A (axial)	$4/\pi = 1.2732$	1.2762	−0.23%
f_π	$180\, m_e = 91.98$ MeV	92.21 MeV	−0.25%
$g_{\pi NN}$	$2\pi^4/15 = 12.99$	~ 12.98	+0.06%
$g^2/(4\pi)$	$\pi^7/225 = 13.42$	~ 13.5	−0.6%
	Neutrino masses		
m_ν (atm.)	$\alpha^3 m_e/4 = 0.049$ eV	0.050 eV	−0.8%
	Quark masses and Higgs		
m_s/m_d	$2\pi^2 = 19.74$	20.0	−1.3%
m_c/m_u	$6\pi^4 = 584$	588	−0.6%
m_b/m_s	$9\pi^2/2 = 44.4$	44.8	−0.8%
m_t/m_c	$14\pi^2 = 138$	136	+1.6%
v (Higgs VEV)	$504\pi^6 m_e = 247.6$ GeV	246.2 GeV	+0.6%
m_H	$v\sqrt{\pi/12} = 126.7$ GeV	125.25 GeV	+1.1%
	Nuclear and hadronic		
R_p (charge radius)	$\hbar c/(C\varphi_{\rm gold}) = 0.865$ fm	0.875 fm	−1.1%
N_c (colors)	$(2\ell+1) = 3$	3	exact
$\mathfrak{su}(3)$ Casimir C_2	$4/3$	$4/3$	$< 10^{-16}$
$\mathfrak{su}(3)$ adjoint C_A	$f^{acd} f^{bcd}/\delta^{ab}$	3	$< 10^{-6}$
QCD b_0 (pure glue)	$[(2\vert\kappa_{\max}\vert+1)+(2j+1)^2](2\ell+1)/3$	11	exact
QCD b_0 ($N_f = 6$)	$11 - (2/3)\times 6$	7	exact
B_d (deuteron)	$C/63 = 2.237$ MeV	2.224 MeV	+0.61%
$B(^3\text{He})$	$C/18 = 7.831$ MeV	7.718 MeV	+1.47%
$B(^4\text{He})$	$C/5 = 28.19$ MeV	28.30 MeV	−0.37%
B(Li-7)	$C \times 5/18 = 39.16$ MeV	39.24 MeV	−0.22%
	Light element abundances		
Y(He-4) floor	$(r_b/r_*)^3 = \sqrt{5} - 2 = 0.236$	0.245 ± 0.003	−3.6%
D/H	$\alpha/(9\pi^3) = 2.62\times 10^{-5}$	2.53×10^{-5}	+3.5%
Li-7/H	Destruction $\gg$ production	1.6×10^{-10}	resolved
τ_n	From G_F, g_A, Q	878.4 s	+8.6%
	Cosmological		
SNe (1701)	$d_L = r\, e^\phi$	Pantheon+	0.166 mag RMS
CMB EE (SPT-3G D1 [39])	0 free params	vs. ΛCDM (6 params)	9.3% vs. 12.6%
BAO (8 surveys)	D_M/r_d, native r_d, 0 free	BOSS/DESI	3.8% RMS
$c^2 = 2GM/r_*$	Machian closure	c observed	0.0000%
Ω_b (UDT)[§]	$\rho_b/\rho_{\rm Mach} \approx 1$	all-baryon	—
$r_{*,\rm cosmo}$	9.164 Gpc (derived)	—	—

[§]The Machian critical density $\rho_{\rm Mach} = 3c^2/(8\pi G r_*^2)$ differs from the ΛCDM critical density. The UDT universe is baryon-dominated relative to its own closure condition; the CMB $\Omega_b = 0.049$ constraint is ΛCDM-framework-specific (baryon loading encodes peak ratios; in UDT the peak ratios are set by the Weinberg angle).

TABLE XIII. Decisive experimental tests. Each row identifies a prediction, the measurement that would strongly disfavor or confirm it, and the relevant experiment. For binary predictions (B-modes, dark matter particle), a positive detection would be incompatible with the framework.

Prediction	Decisive test	Experiment
$m_p/m_e = 6\pi^5$	Ratio deviates $> 0.01\%$ from $6\pi^5$	CODATA mass measurements
$m_\mu/m_e = 20\pi^3/3$	Ratio deviates $> 0.1\%$	Muon $g-2$ (Fermilab, J-PARC)
$m_\pi = 84\pi\, m_e$	Ratio deviates $> 0.5\%$	PIENU, precision pion mass
$1/\alpha_{\rm EM} = 137.036$	Deviation $> 0.01\%$	Electron $g-2$ (Harvard)
Δm^2_{21} (MSW-free)	Reactor value inconsistent with 6.98×10^{-5} eV^2 at $> 3\sigma$	JUNO (0.5%, matter-free)[‡]
$\sin\theta_C = 9/40$	Deviation $> 0.5\%$	LHCb CKM measurements
$v = 504\pi^6 m_e$	VEV $\neq 247.6$ GeV at $> 2\%$	LHC Run 3, HL-LHC
$\sin^2\theta_{12} = 4/13$	Inconsistent at $> 3\sigma$	JUNO, DUNE
$\sin^2\theta_{23} = 4/7$	Inconsistent at $> 3\sigma$	Hyper-Kamiokande, DUNE
$\sin^2\theta_{13} = 1/45$	Inconsistent at $> 3\sigma$	JUNO
$m_\nu = \alpha^3 m_e/4$	Direct mass outside 0.04–0.06 eV	KATRIN, Project 8
$m_n - m_p = (5/4)\alpha_{\rm EM} C$	Deviation $> 2\%$	Independent numerical verification
Static universe: drift	Redshift drift $dz/dt \neq 0$ at $> 5\sigma$	Sandage–Loeb (ELT, 2035+)[†]
Time dilation	Width $\neq (1+z)$ at $> 5\sigma$	Consistent with existing data[†]
Galaxies at all z	No mature galaxies at $z > 15$	JWST, Roman Space Telescope
No B-mode polarization	B-modes [56] detected at $r > 0.001$	CMB-S4, LiteBIRD
Baryonic halo: microlensing	$\tau_{\rm LMC}$ inconsistent with NS+BH halo	Roman Space Telescope[‖]
No dark matter particle	WIMP or axion detected	LZ, XENONnT, ADMX
$c^2 = 2GM/r_*$	Machian relation deviates $> 1\%$	BAO + CMB joint analysis
GW waveform	$O(\phi^2)$ merger correction absent at $> 3\sigma$	Cosmic Explorer, ET (2035+)

[†]The SN time-dilation and Tolman surface-brightness tests are satisfied by the metric (1) and do not constitute evidence against the static interpretation; see Sec. IV.

[‡]The NuFIT global fit value 7.42×10^{-5} eV^2 incorporates MSW corrections absent in UDT. The approximately MSW-independent KamLAND reactor value is 7.49×10^{-5} eV^2. JUNO will provide the definitive MSW-free measurement; see Sec. XV B.

[‖]UDT predicts the LMC microlensing optical depth is dominated by NS and BH halo populations ($f \sim 2\%$ of total halo), with WDs remaining in the disk. Roman wide-field microlensing surveys will test the predicted NS+BH halo fraction and mass function.

TABLE XIV. Joint cosmological fit at the derived $\mu_g = 0.247$ Gpc^{-1} (zero fitted cosmological parameters). SNe uses $d_L = r(1+z)$ with M marginalized analytically. BAO compares transverse D_M/r_d (Sec. VI). CMB uses WKB2 peaks with Diophantine $q_1 = 2/9$, $q_2 = -1/20$, peak 1 anchored.

Dataset	N	Observable	RMS
SNe (Pantheon+)	1590	$\mu(z)$	0.164 mag
BAO (8 surveys)	8	D_M/r_d	3.8%
CMB peaks (Planck)	7	ℓ_n	1.35%